This is CS50

CS50
STRESS BALL
CS50
STRESS BALL

CS50
STRESS BALL

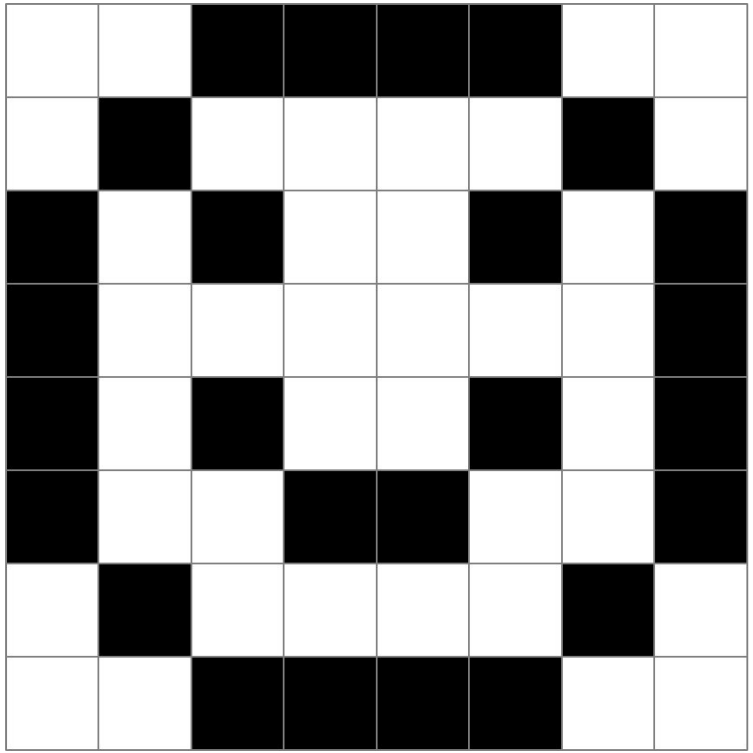
CS50
STRESS BALL

550

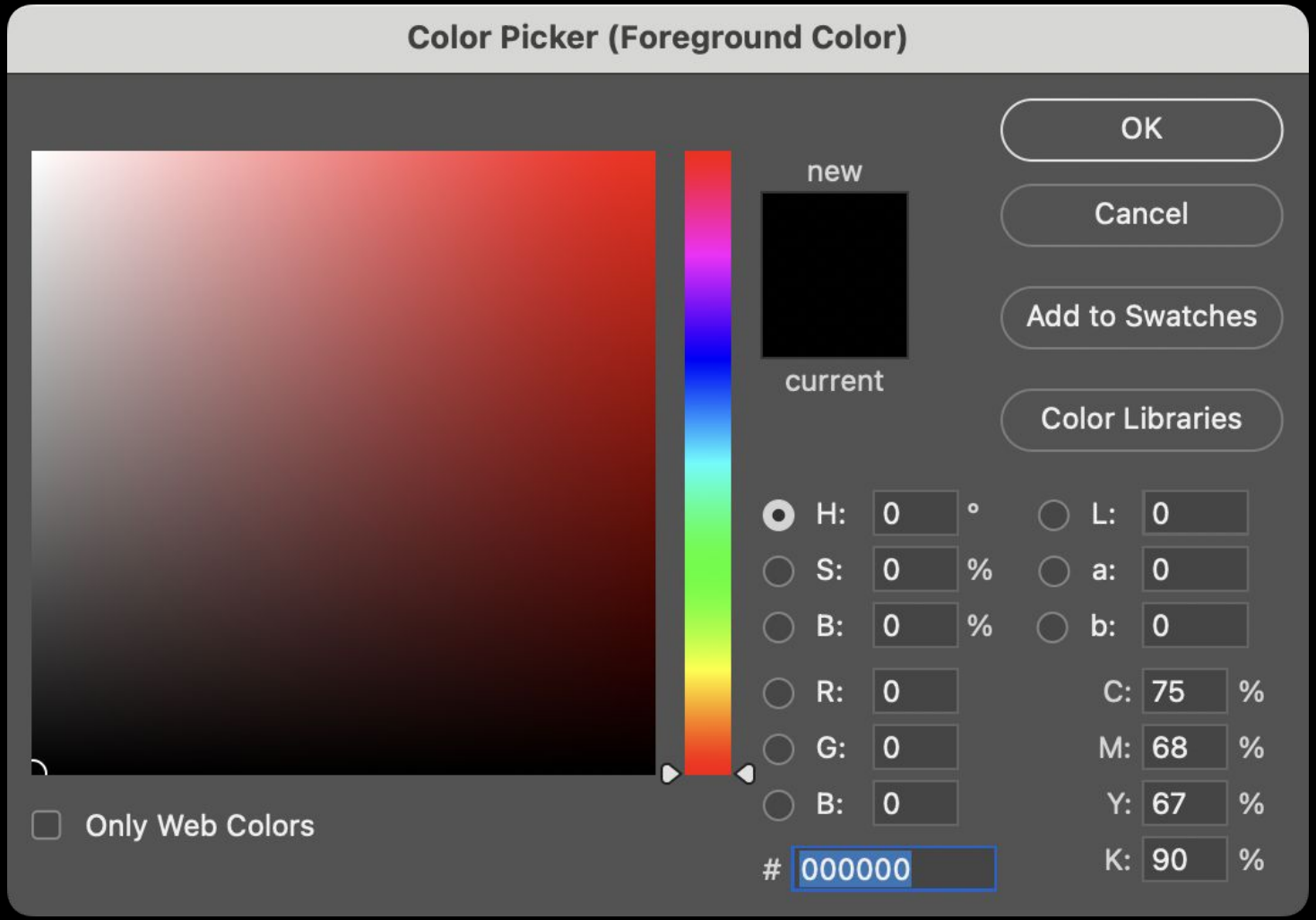
```
1 1 0 0 0 0 1 1
1 0 1 1 1 1 0 1
0 1 0 1 1 0 1 0
0 1 1 1 1 1 1 0
0 1 0 1 1 0 1 0
0 1 1 0 0 1 1 0
1 0 1 1 1 1 0 1
1 1 0 0 0 0 1 1
```

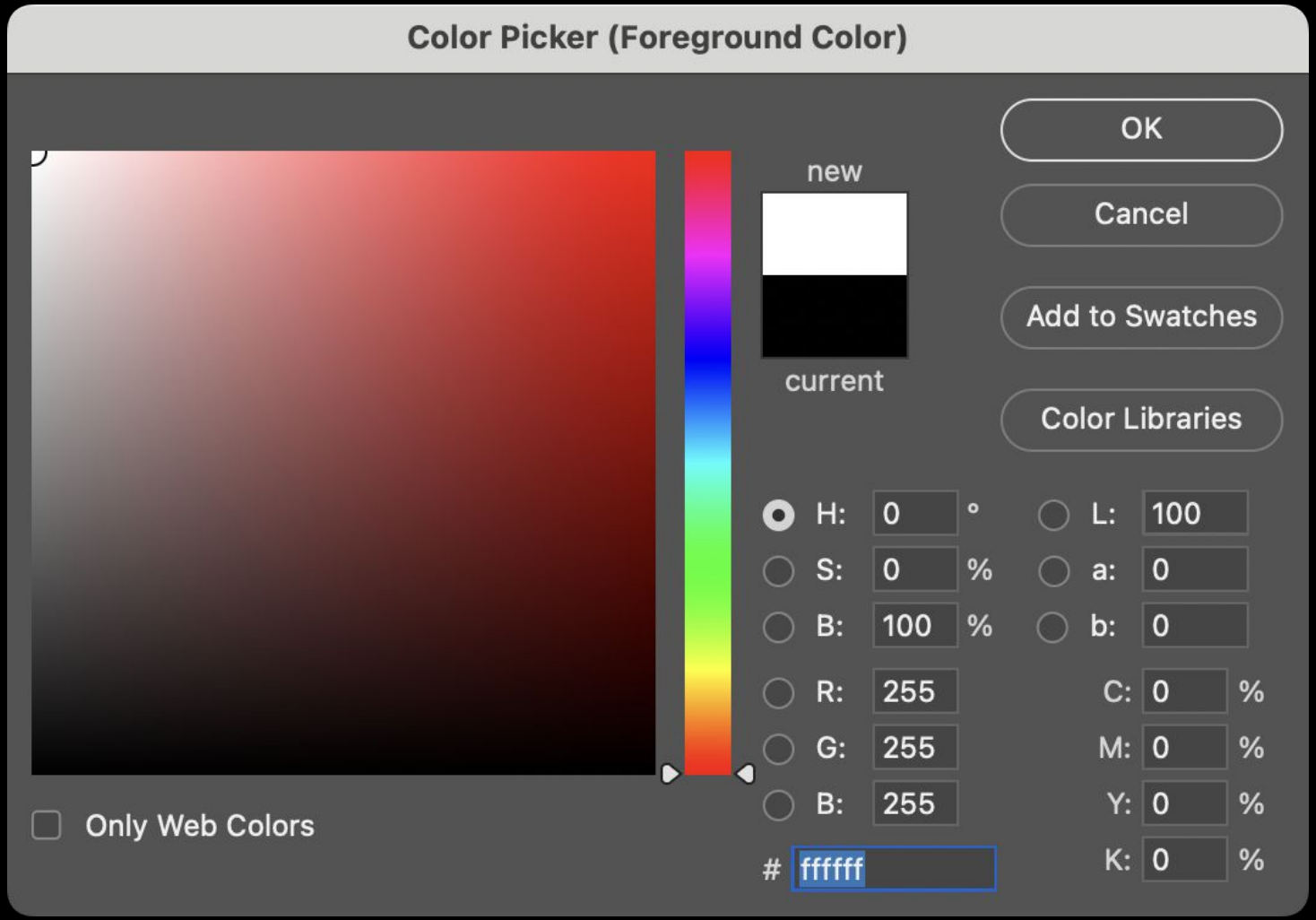
```
    0 0 0 0
  0         0
0   0     0   0
0             0
0   0     0   0
0     0 0     0
  0         0
    0 0 0 0
```

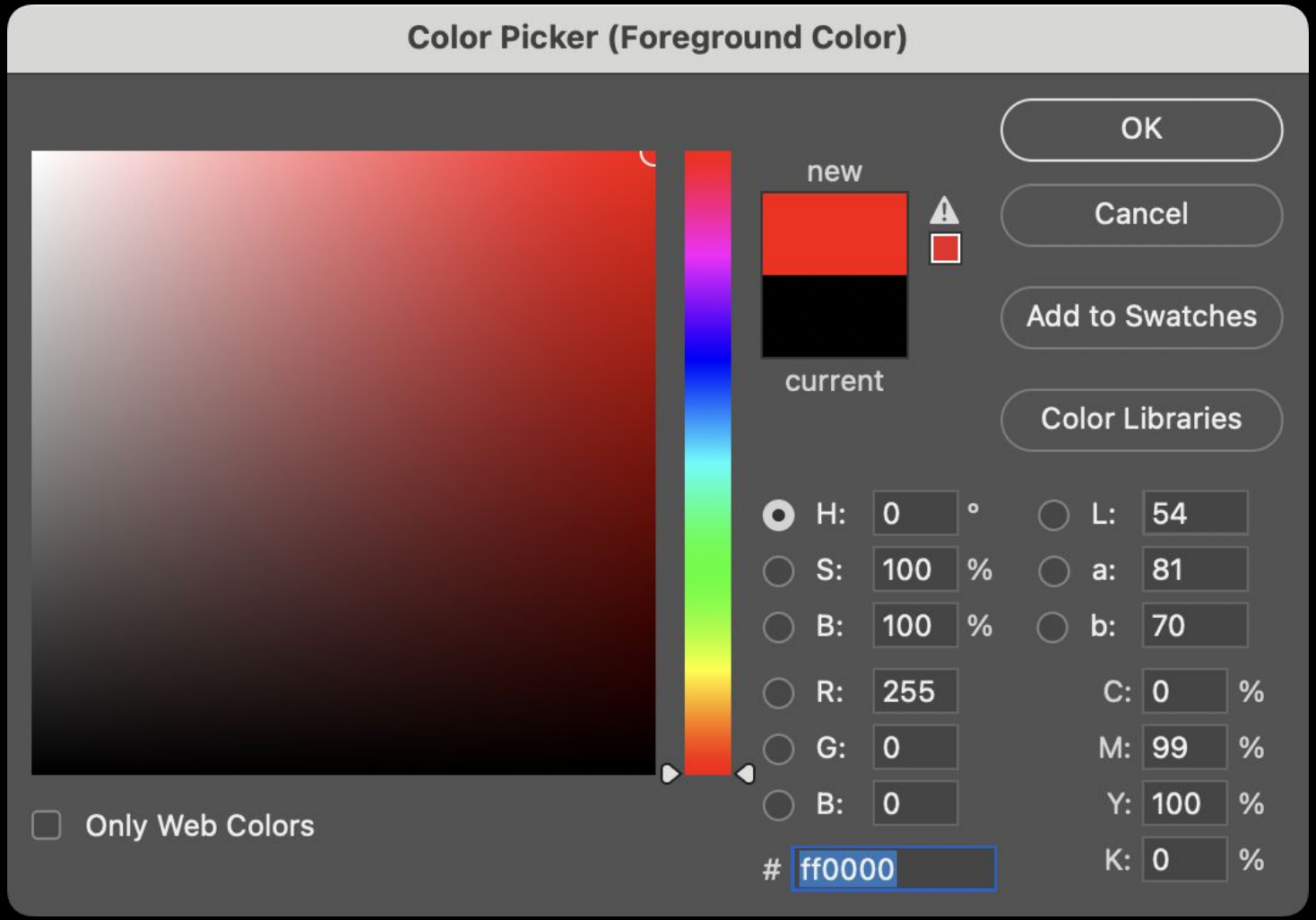
```
1 1 0 0 0 0 1 1
1 0 1 1 1 1 0 1
0 1 0 1 1 0 1 0
0 1 1 1 1 1 1 0
0 1 0 1 1 0 1 0
0 1 1 0 0 1 1 0
1 0 1 1 1 1 0 1
1 1 0 0 0 0 1 1
```

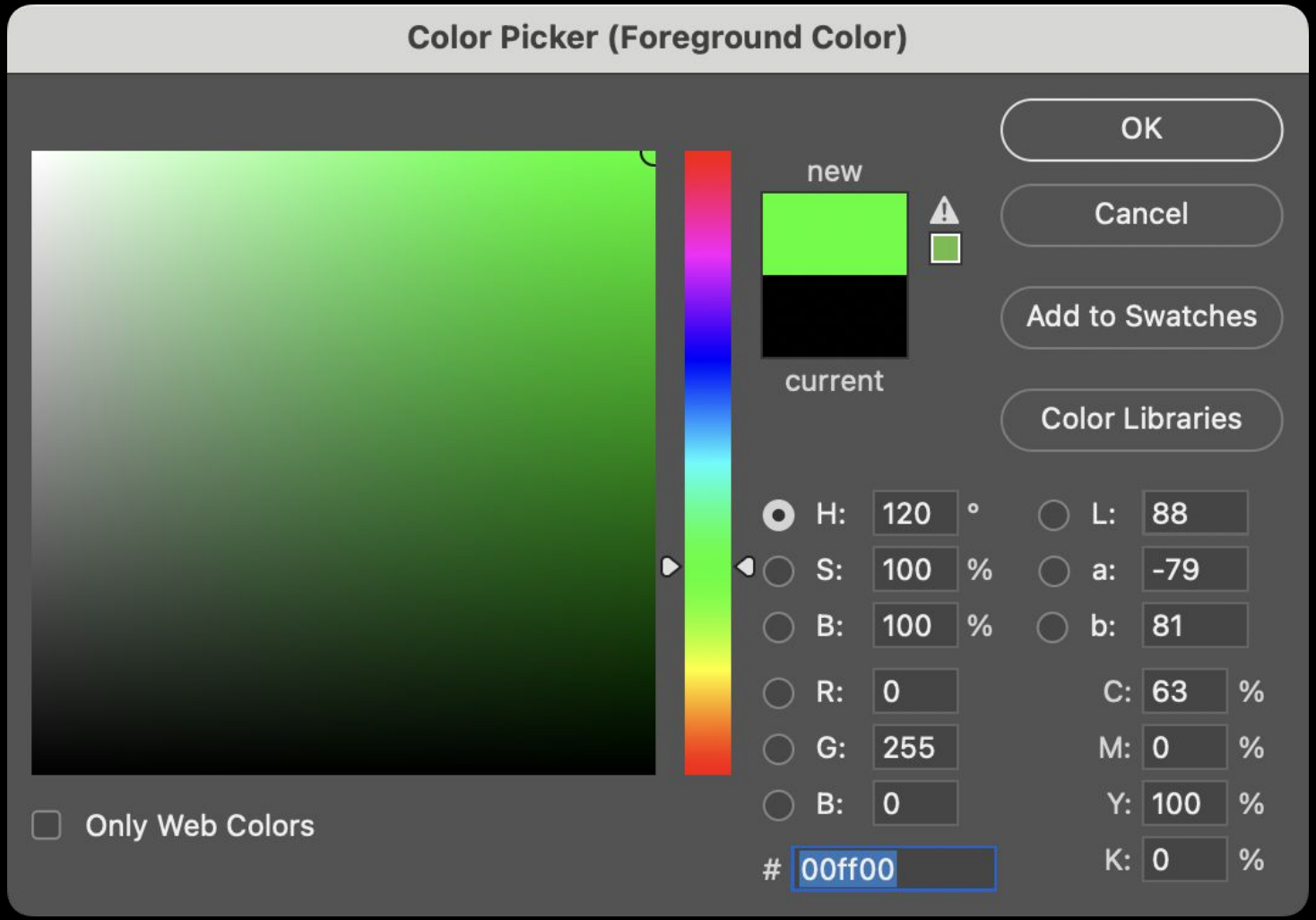


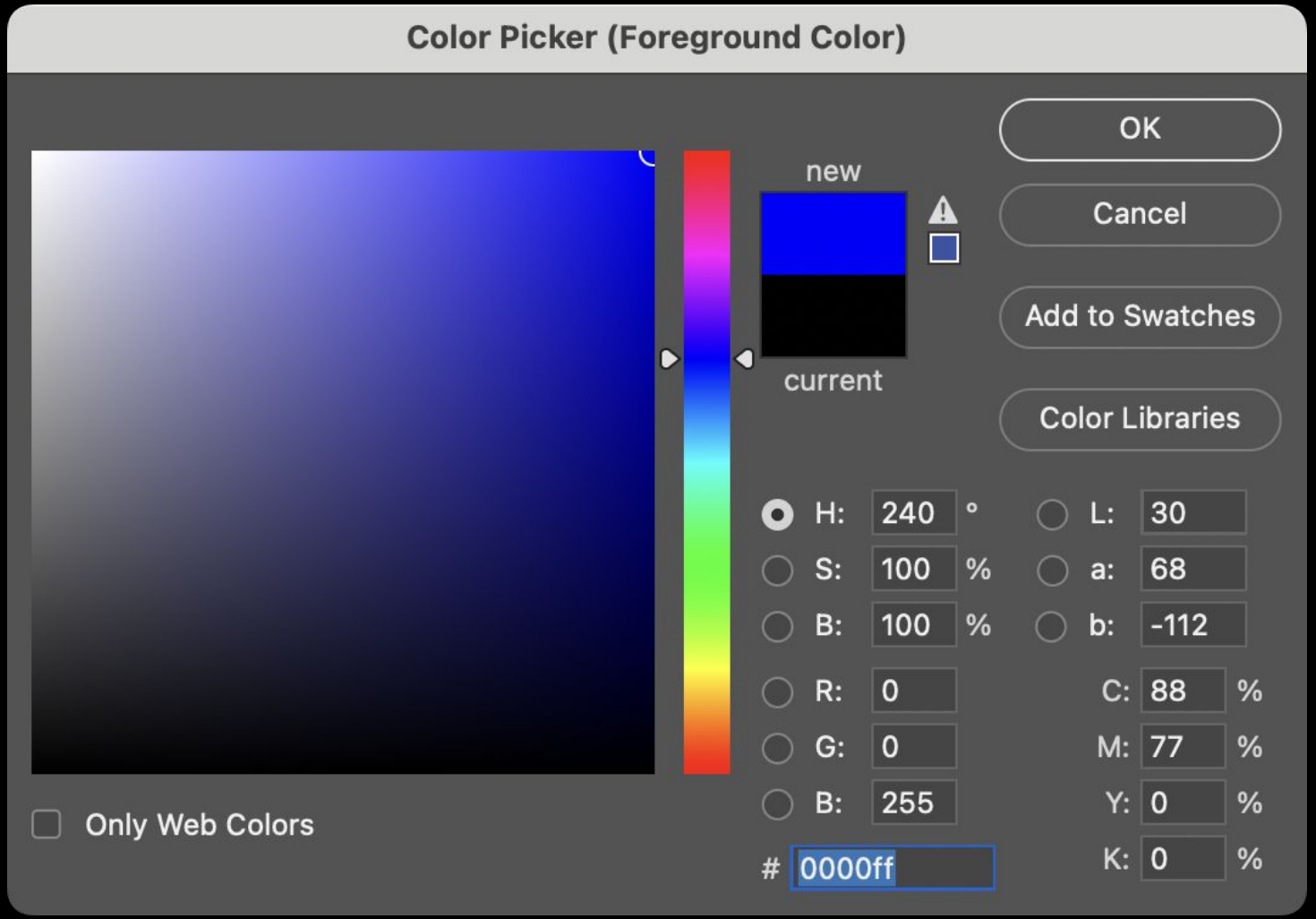
RGB


Color Picker (Foreground Color)
OK
Cancel
Add to Swatches
Color Libraries
new
current
H: 0 °
S: 0 %
B: 0 %
R: 0
G: 0
B: 0
000000
L: 0
a: 0
b: 0
C: 75 %
M: 68 %
Y: 67 %
K: 90 %
Only Web Colors


Color Picker (Foreground Color)
OK
Cancel
Add to Swatches
Color Libraries
new
current
H: 0 °
S: 0 %
B: 100 %
R: 255
G: 255
B: 255
ffffff
L: 100
a: 0
b: 0
C: 0 %
M: 0 %
Y: 0 %
K: 0 %
Only Web Colors


Color Picker (Foreground Color)
OK
Cancel
Add to Swatches
Color Libraries
new
current
H: 0 °
S: 100 %
B: 100 %
R: 255
G: 0
B: 0
ff0000
L: 54
a: 81
b: 70
C: 0 %
M: 99 %
Y: 100 %
K: 0 %
Only Web Colors


Color Picker (Foreground Color)
OK
Cancel
Add to Swatches
Color Libraries
new
current
H: 120 °
S: 100 %
B: 100 %
R: 0
G: 255
B: 0
00ff00
L: 88
a: -79
b: 81
C: 63 %
M: 0 %
Y: 100 %
K: 0 %
Only Web Colors


Color Picker (Foreground Color)
OK
Cancel
Add to Swatches
Color Libraries
new
current
H: 240 °
S: 100 %
B: 100 %
R: 0
G: 0
B: 255
0000ff
L: 30
a: 68
b: -112
C: 88 %
M: 77 %
Y: 0 %
K: 0 %
Only Web Colors

0 1

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9 a b c d e f

0 1 2 3 4 5 6 7 8 9 A B C D E F

hexadecimal

base-16

16^1 16^0

# #

16 1

# #

16 1

00

16 1

01

16 1

02

16 1

03

16 1

04

16 1

05

16 1

06

16 1

07

16 1

08

16 1

09

16 1

0A

16 1

0B

16 1

0C

16 1

0D

16 1

0E

16 1

0F

16 1

10

16 1

11

16 1

12

16 1

■ ■ ■

16 1

FF

16 1

FF

$16 \times F + 1 \times F$

16 1

FF

$16 \times 15 + 1 \times 15$

16 1

FF

240 + 15

16 1

FF

255

FF

F F

1111 1111

0 1 2 3 4 5 6 7

8 9 10 11 12 13 14 15

0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F

0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F
10	11	12	13	14	15	16	17
18	19	1A	1B	1C	1D	1E	1F

0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7
0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
0x10	0x11	0x12	0x13	0x14	0x15	0x16	0x17
0x18	0x19	0x1A	0x1B	0x1C	0x1D	0x1E	0x1F

```
int n = 50;
```

50

n

50

0x123

&

*

%p

pointers

```
int n = 50;

int *p = &n;
```

```
int n = 50;

int *p = &n;
```

```
int n = 50;

int* p = &n;
```

```
int n = 50;

int * p = &n;
```

```
int n = 50;

int *p = &n;
```

50

n

50

0x123

0x123

p

50

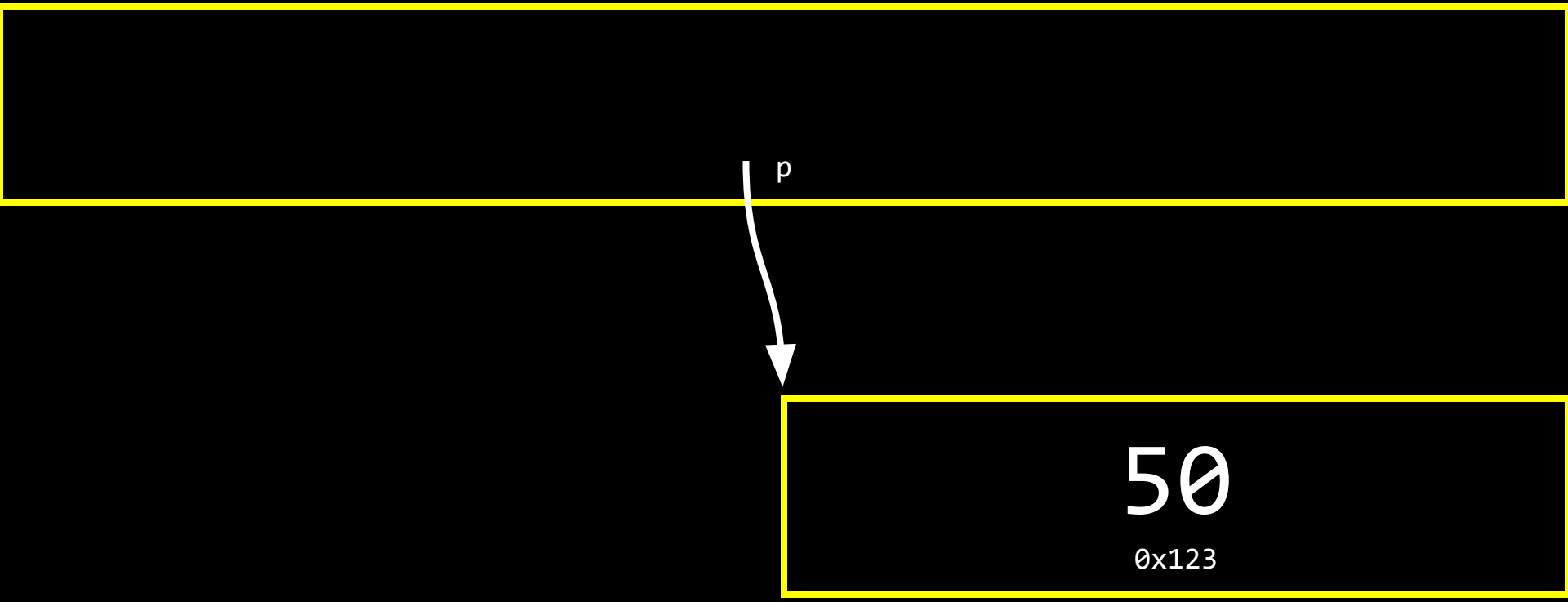
0x123

0x123

p

50

0x123


p
50
0x123

string

```
string s = "HI!";
```

H	I	!	\0

H	I	!	\0
s[0]	s[1]	s[2]	s[3]

H	I	!	\0
0x123	0x124	0x125	0x126

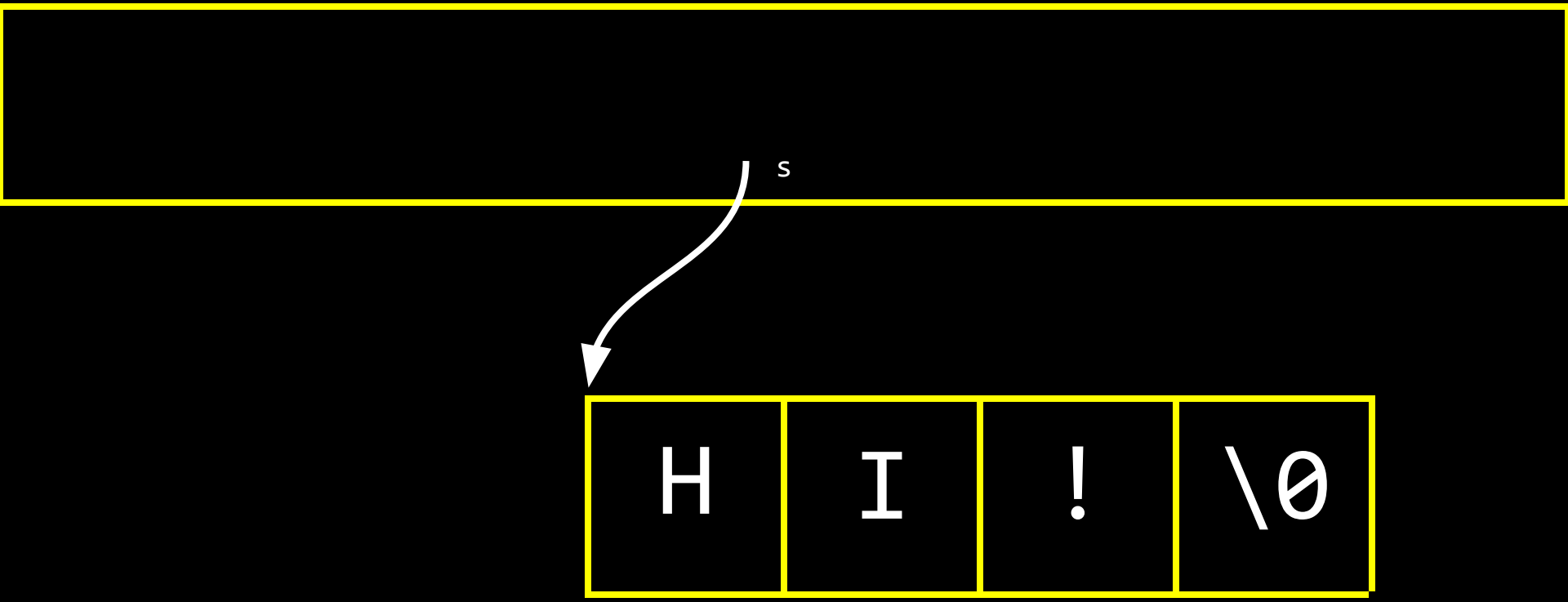
s

H	I	!	\0
0x123	0x124	0x125	0x126

0x123

s

H	I	!	\0
0x123	0x124	0x125	0x126


s
H
I
!
\0

```
string s = "HI!";
```

```
string s = "HI!";
```

```
char *s = "HI!";
```

```
typedef struct
{
    string name;
    string number;
} person;
```

```
typedef struct
{
    string name;
    string number;
} person;
```

```
typedef struct
{
    string name;
    string number;
} person;
```

```
typedef struct
{
    string name;
    string number;
} person;
```

typedef

```
typedef int
```

```
typedef int integer;
```

typedef

```
typedef char *
```

```
typedef char *string;
```

```
typedef char *string;
```

cs50.h

pointer arithmetic

0x123

s

H	I	!	\0
0x123	0x124	0x125	0x126

50
i

50
j

strcmp

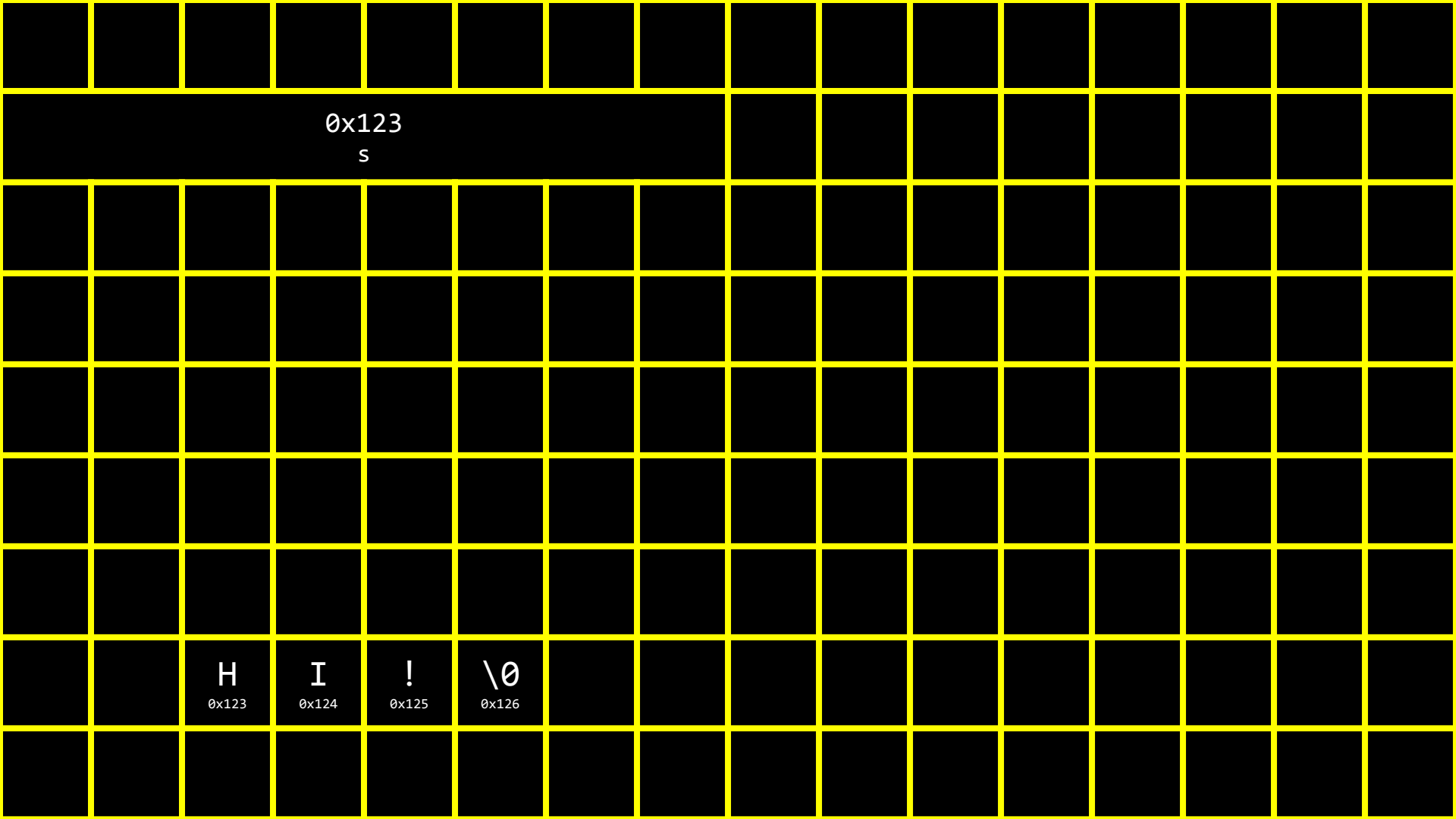
s

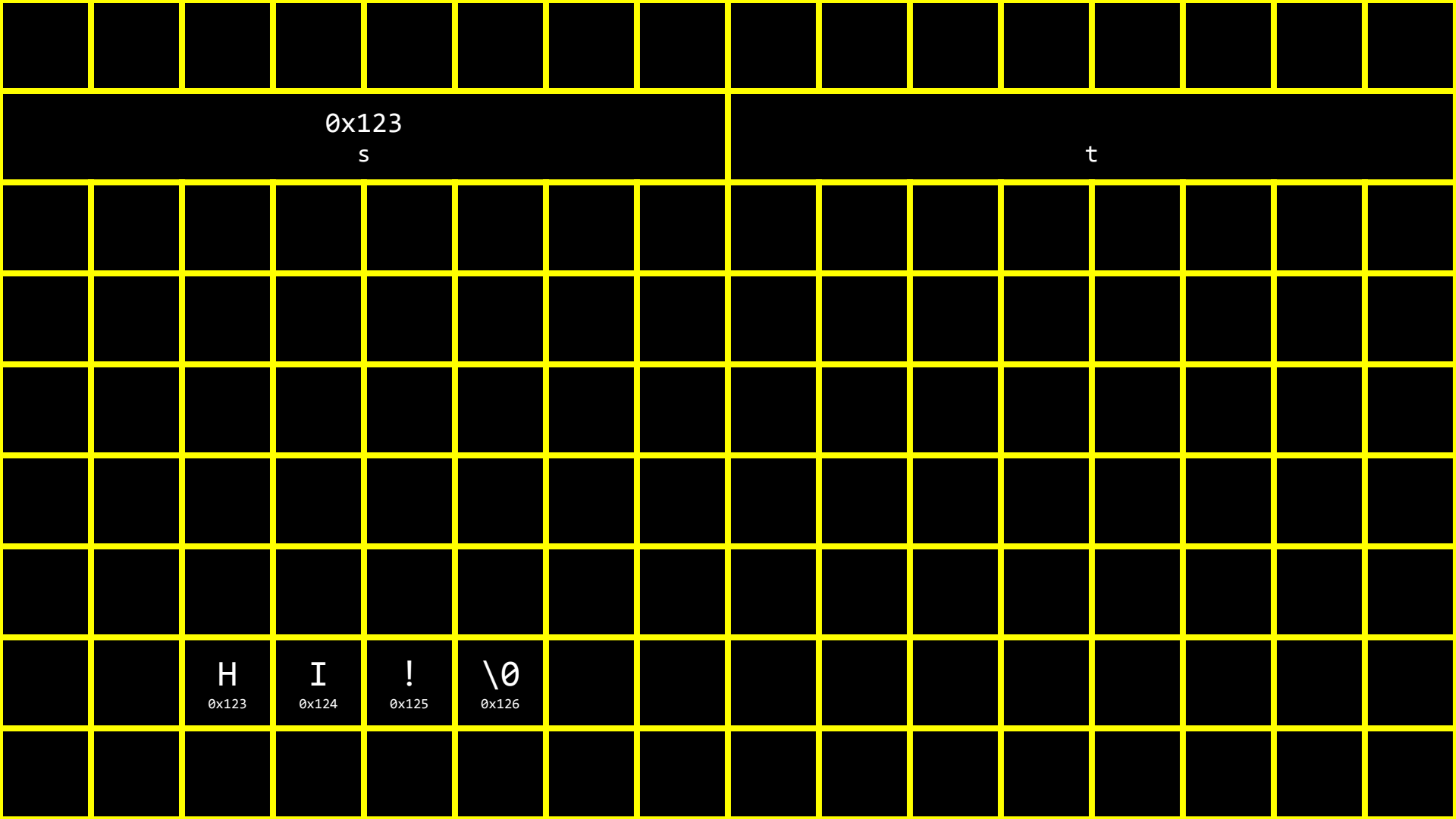
s

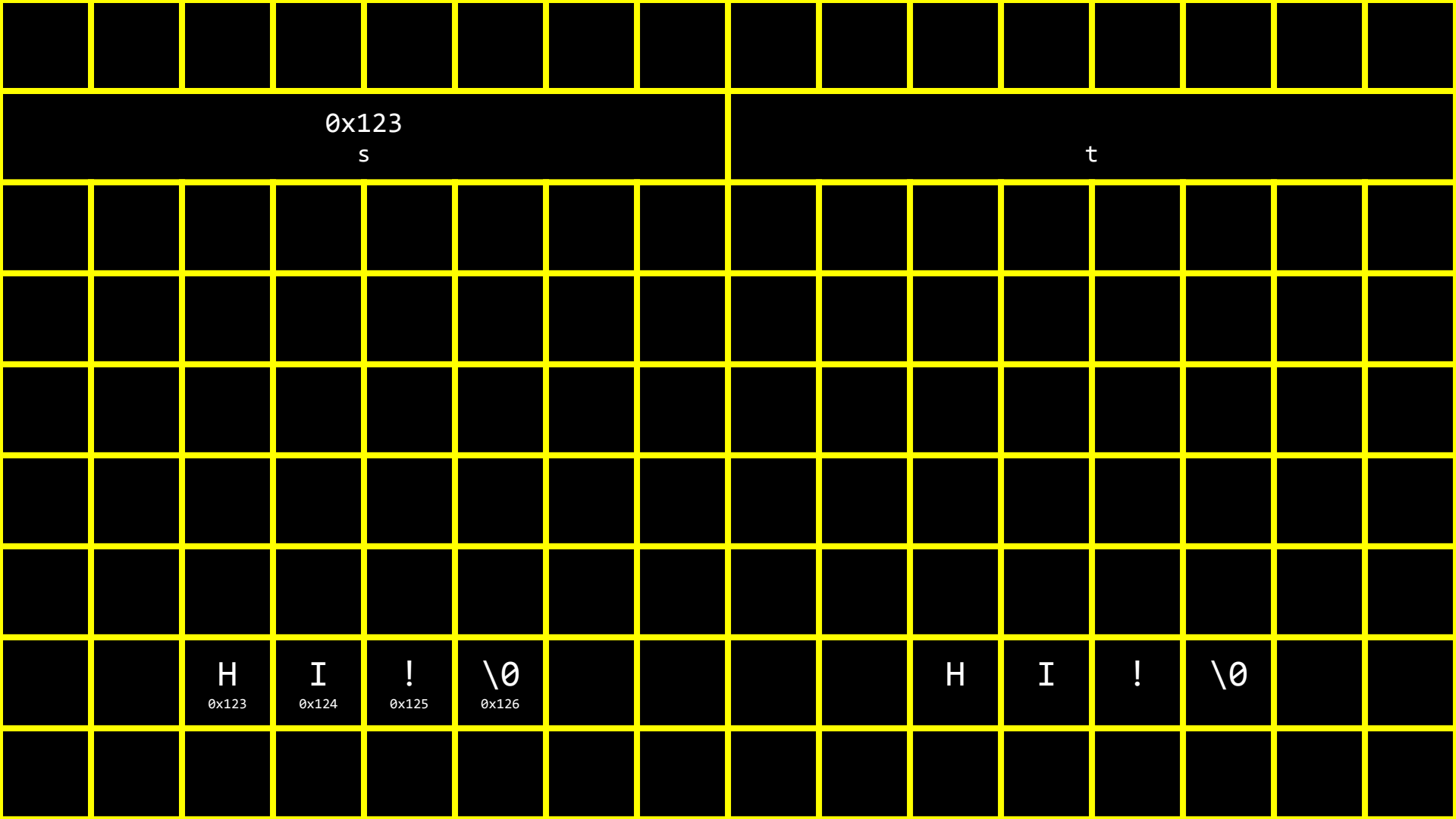
H I ! \0

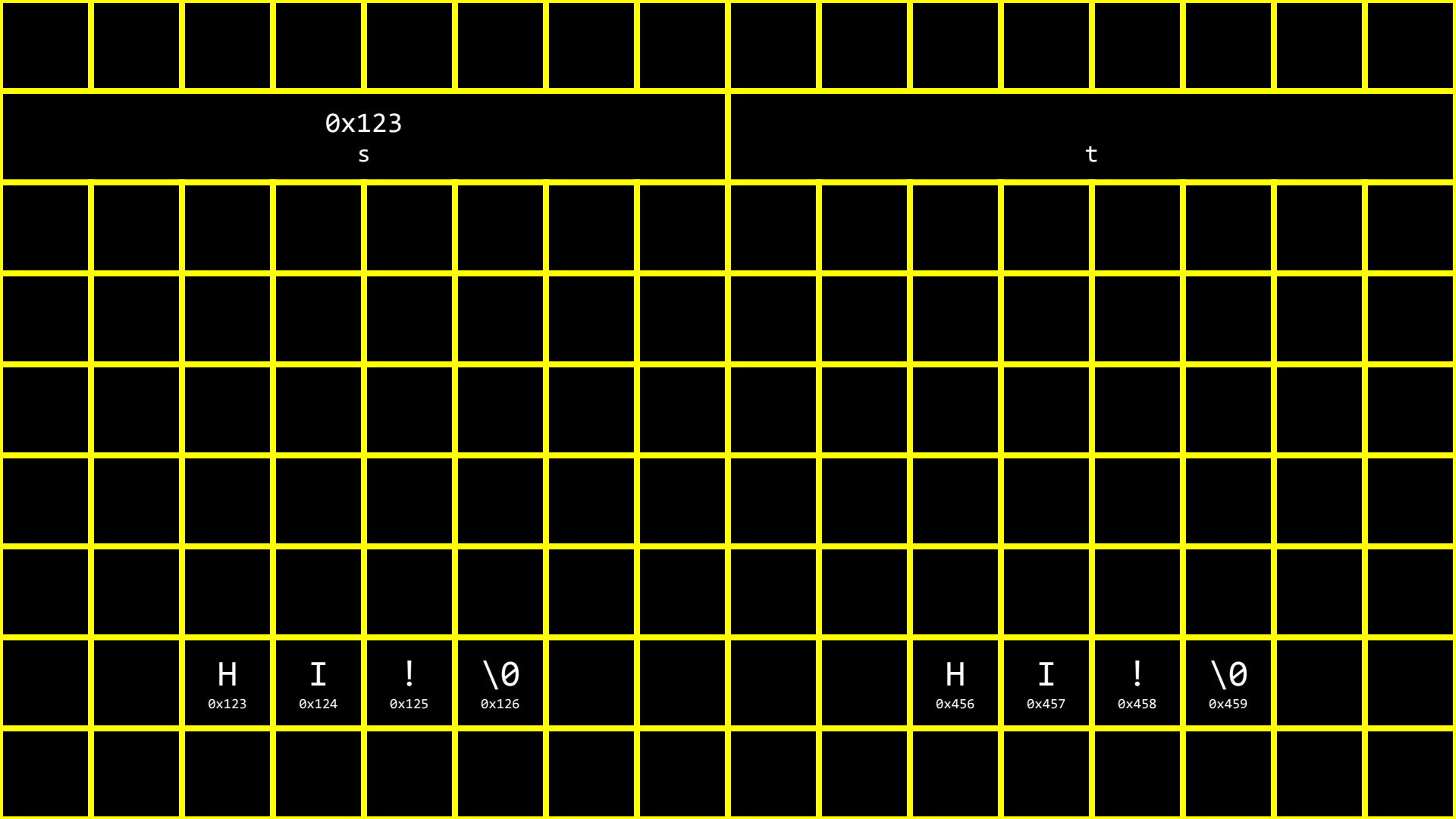
s

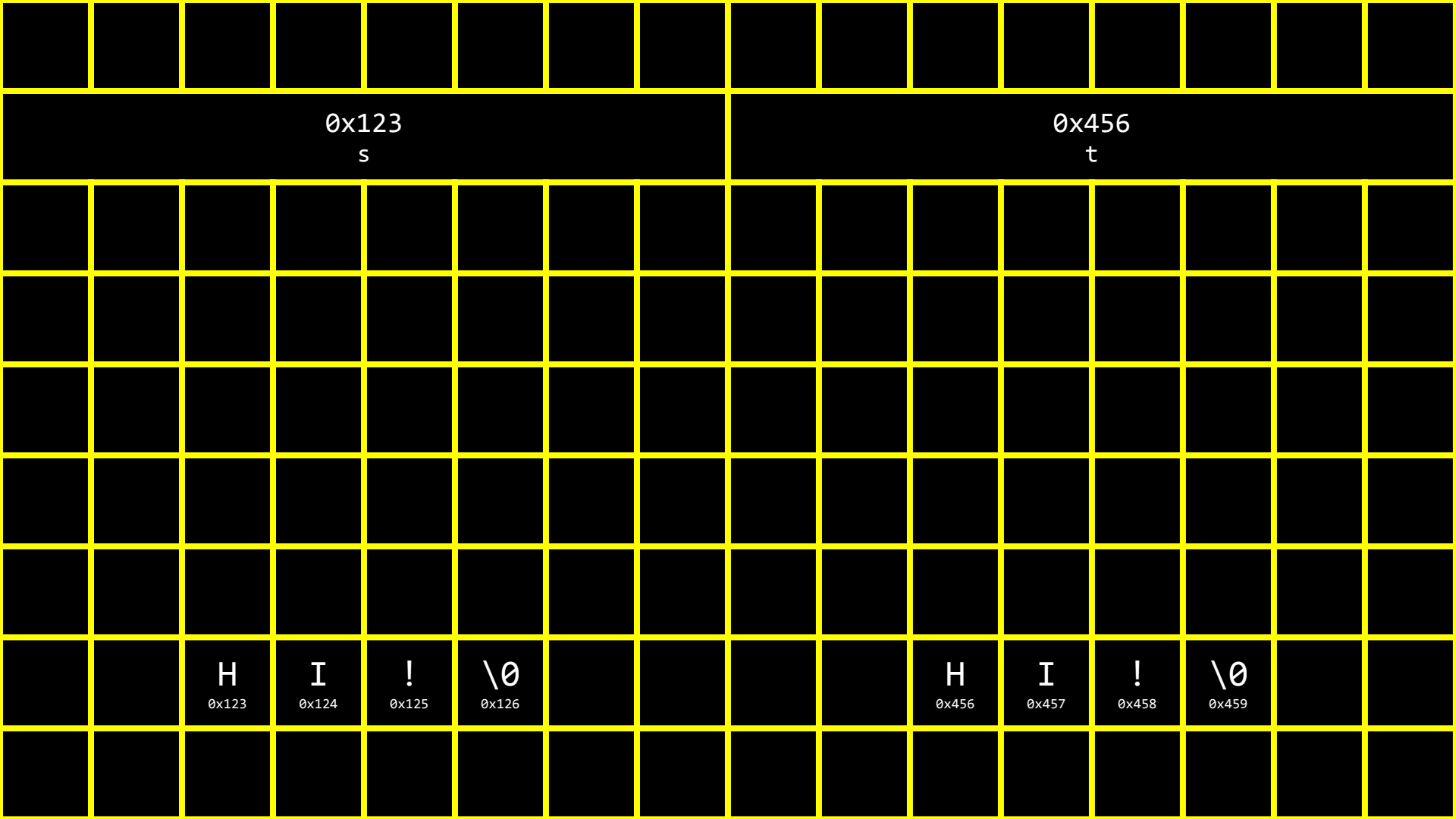
H	I	!	\0
0x123	0x124	0x125	0x126

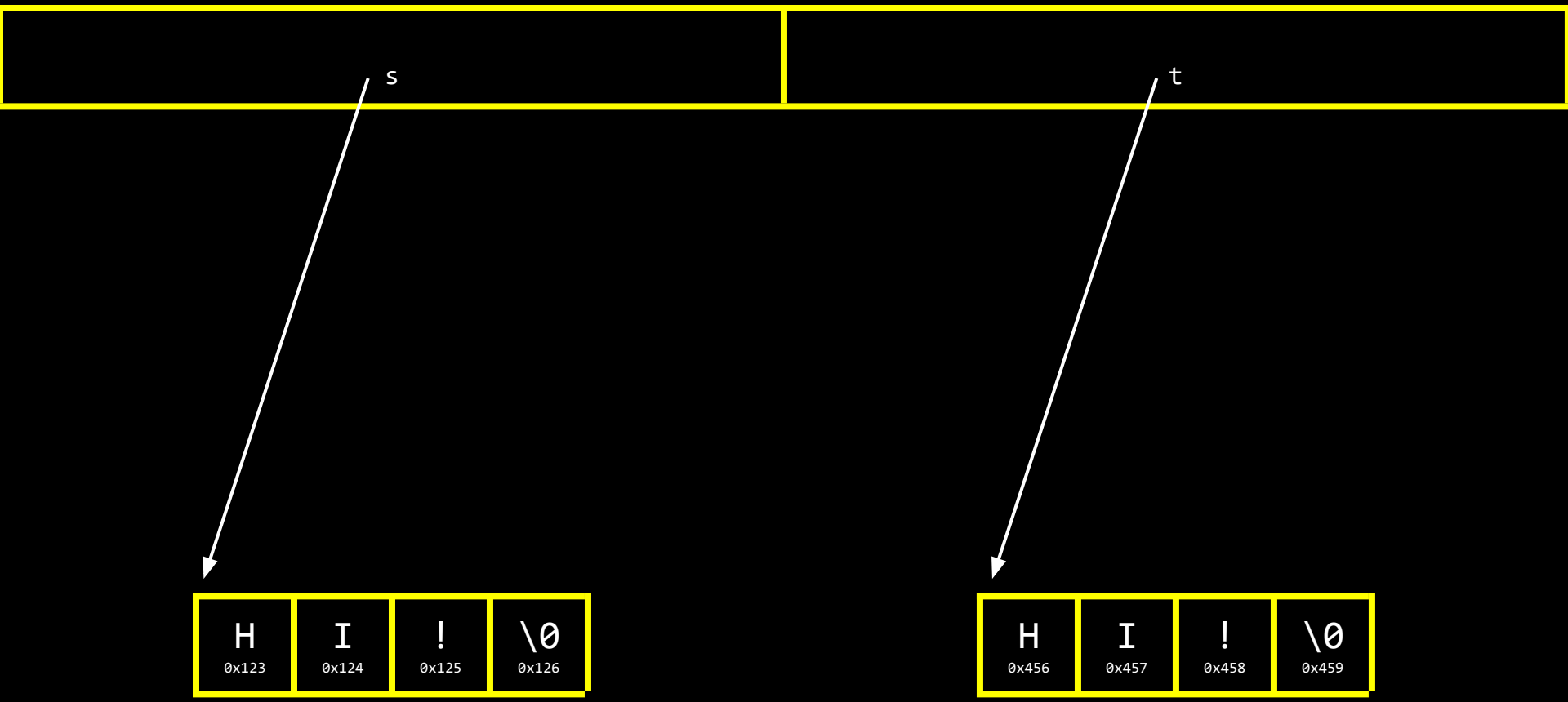

0x123
s
H
0x123
I
0x124
!
0x125
\0
0x126


0x123
s
t
H
0x123
I
0x124
!
0x125
\0
0x126


0x123
s
t
H
0x123
I
0x124
!
0x125
\0
0x126
H
I
!
\0


0x123
s
t
H
0x123
I
0x124
!
0x125
\0
0x126
H
0x456
I
0x457
!
0x458
\0
0x459

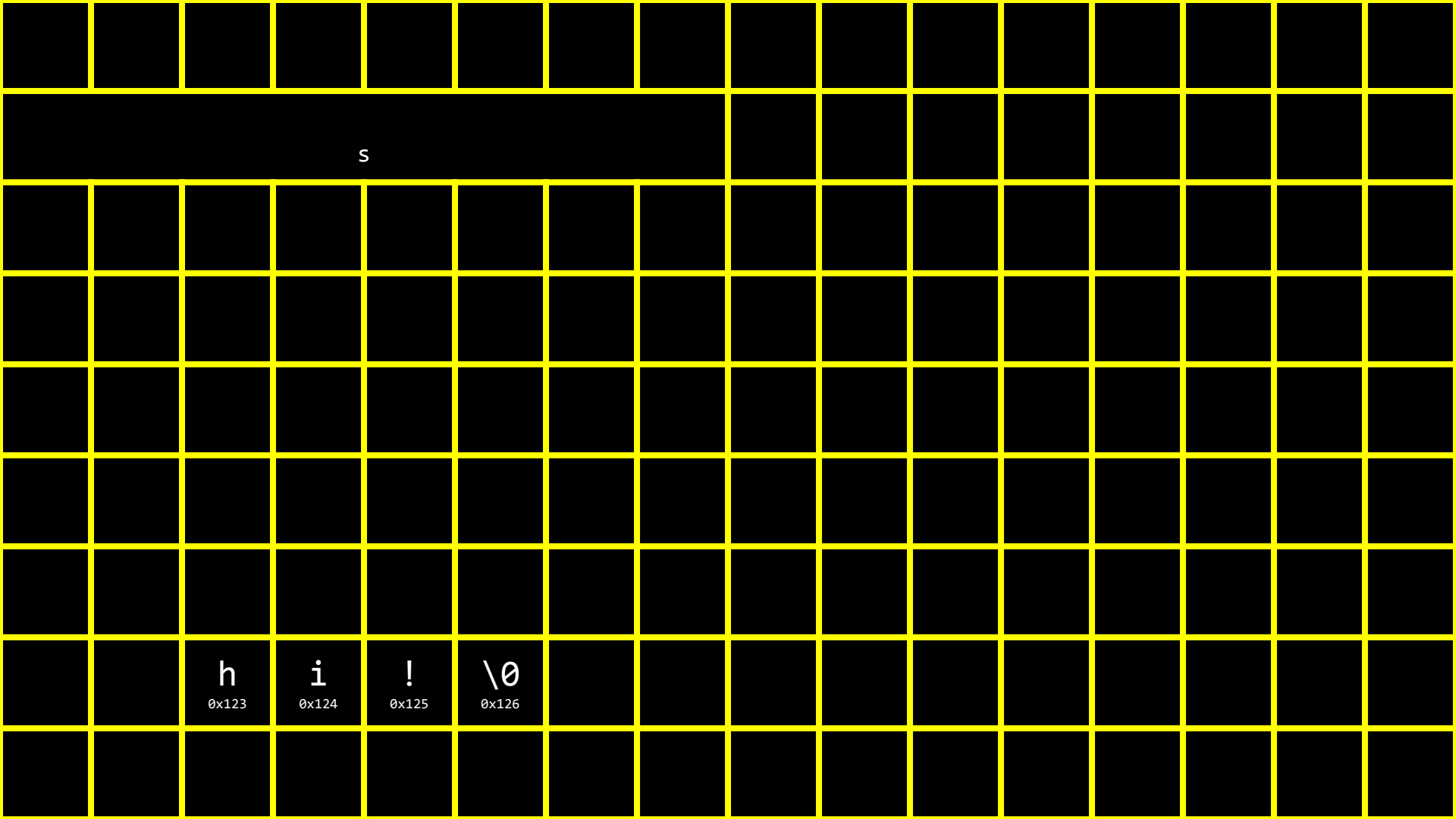

0x123
s
0x456
t
H
0x123
I
0x124
!
0x125
\0
0x126
H
0x456
I
0x457
!
0x458
\0
0x459

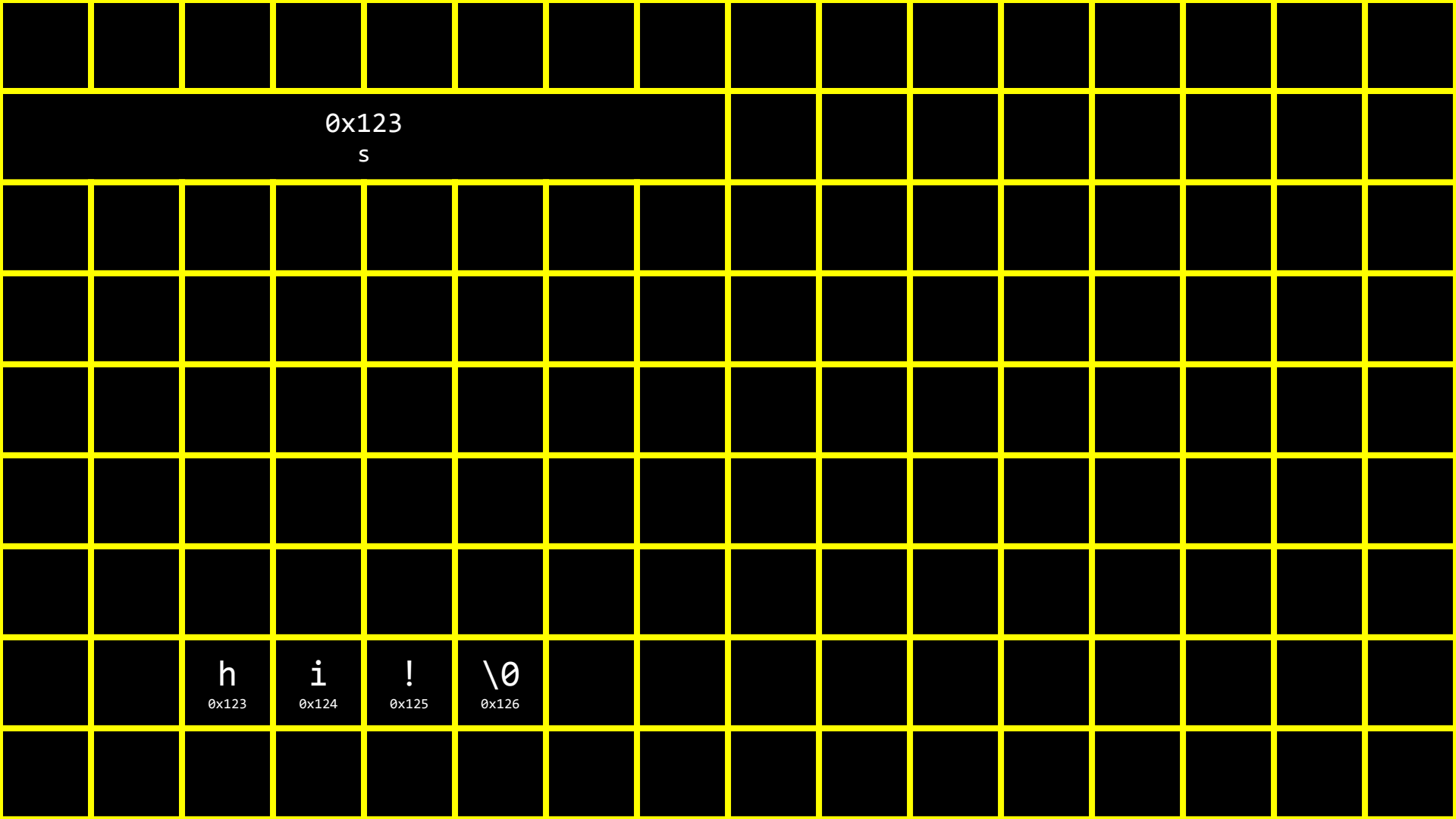

s
t
H
0x123
I
0x124
!
0x125
\0
0x126
H
0x456
I
0x457
!
0x458
\0
0x459

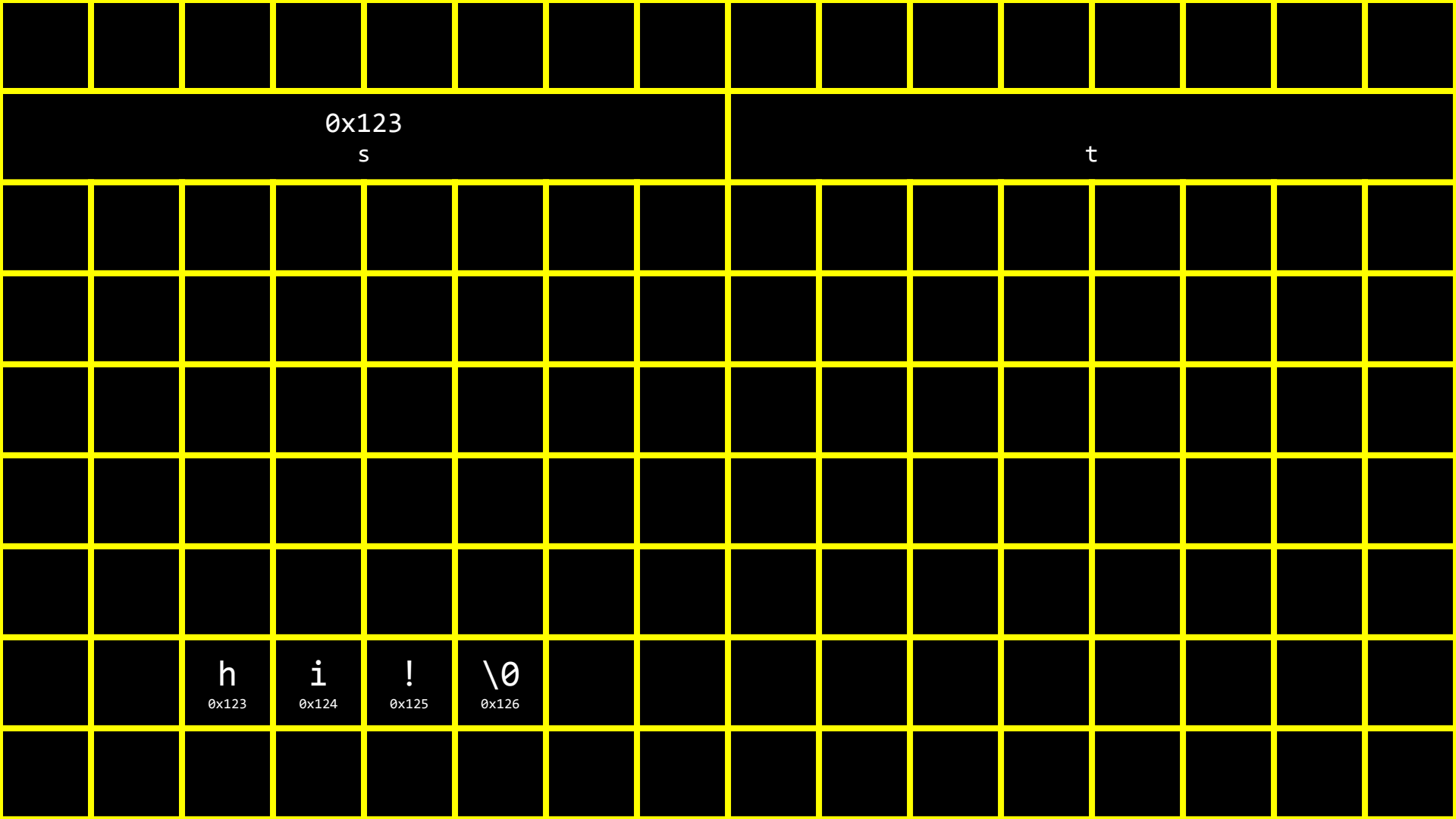
s

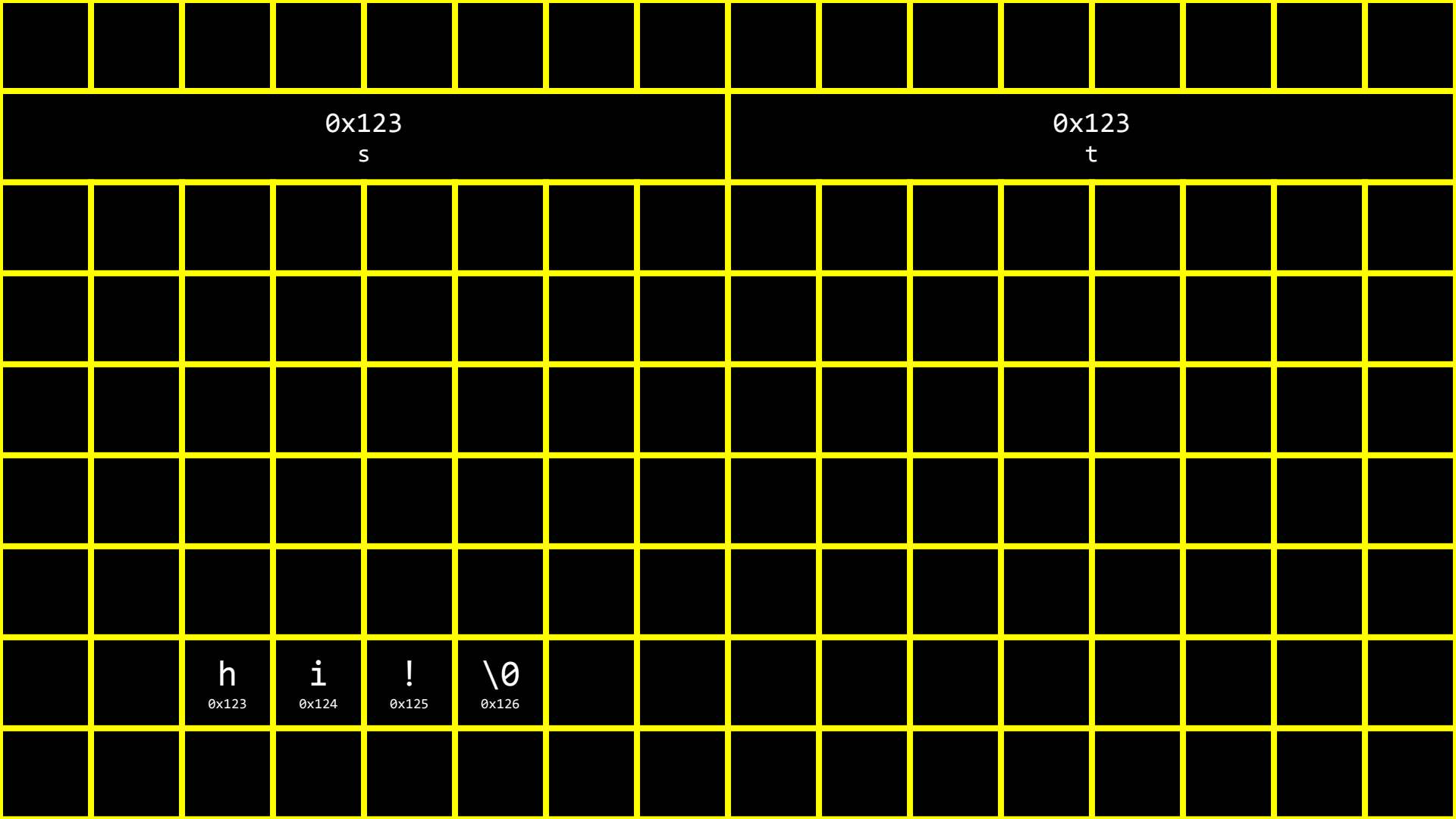
s

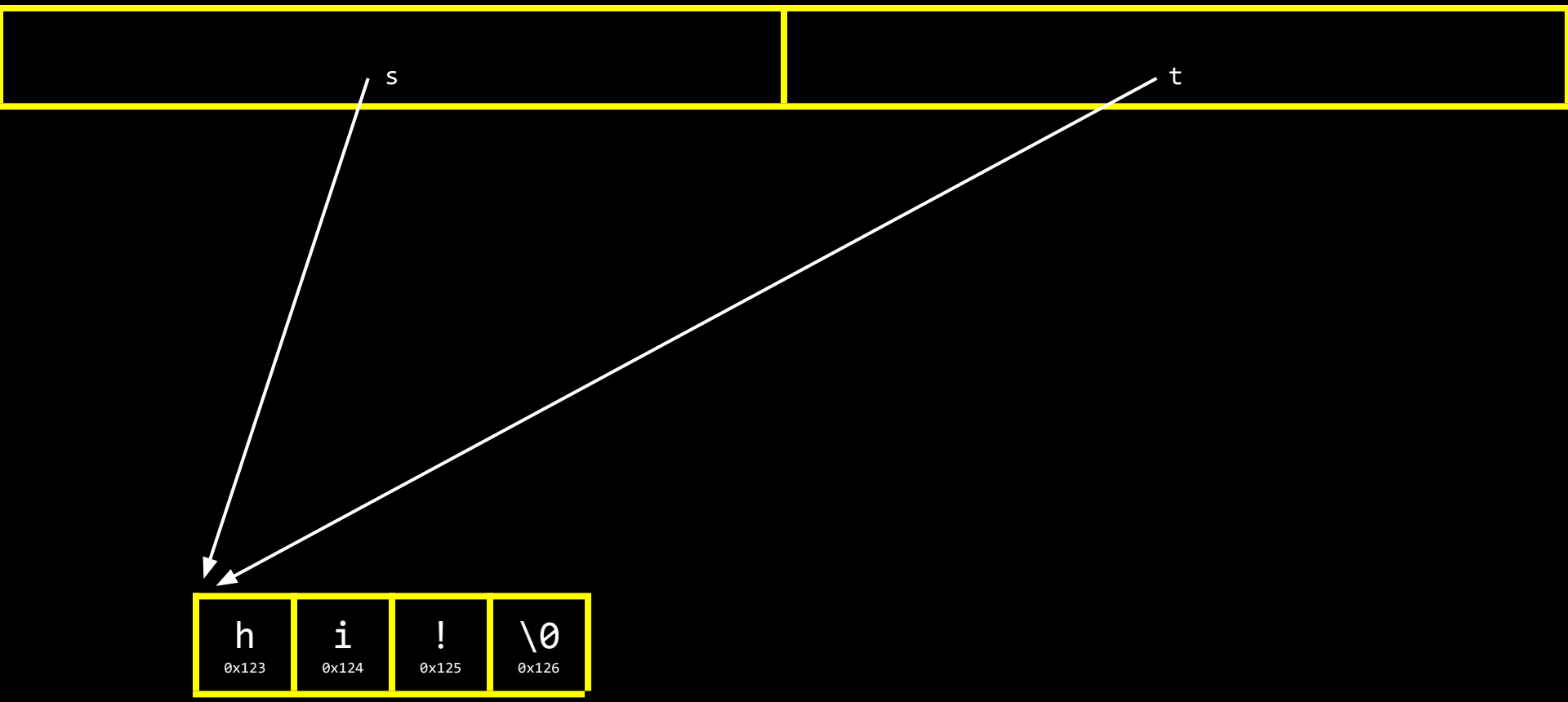
h i ! \0

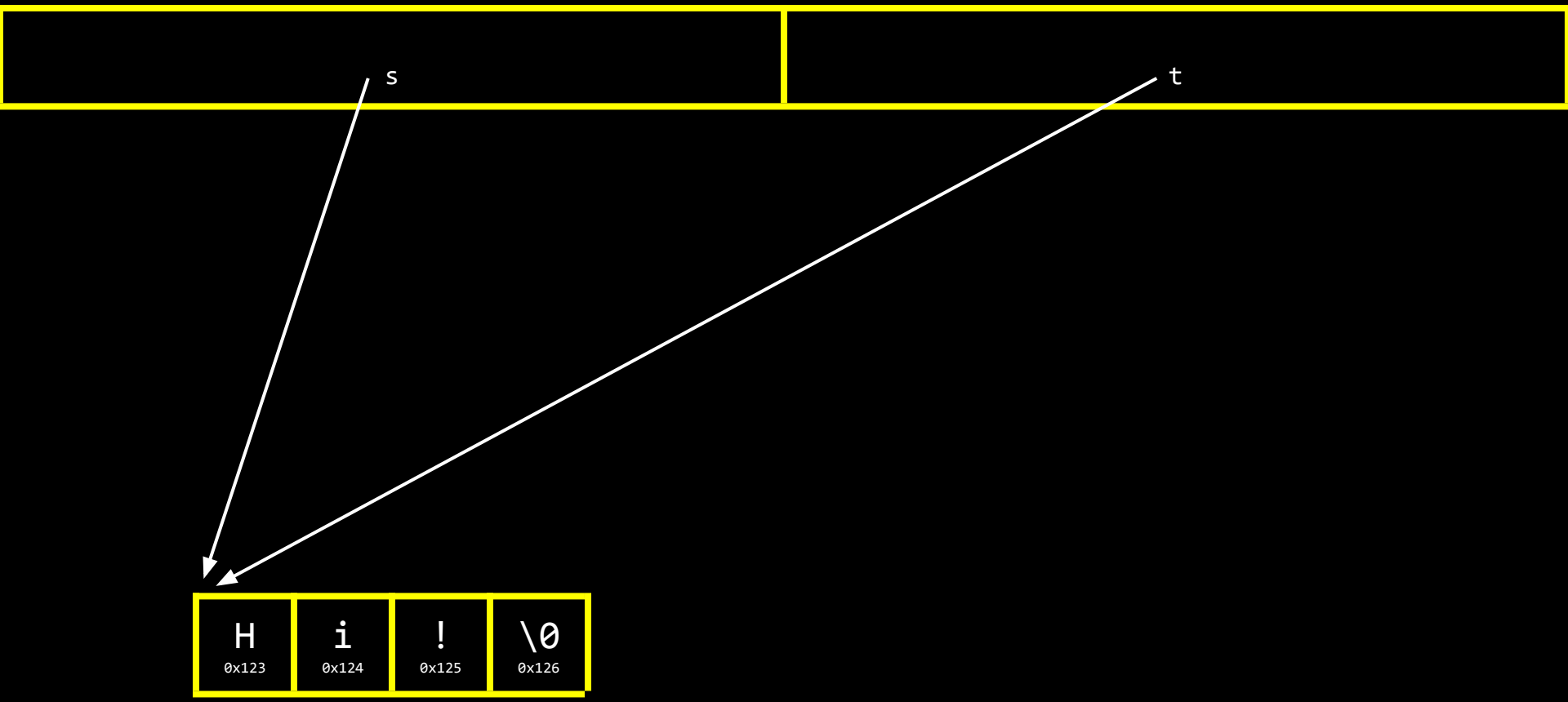

s
h
0x123
i
0x124
!
0x125
\0
0x126


0x123
s
h
0x123
i
0x124
!
0x125
\0
0x126


0x123
s
t
h
0x123
i
0x124
!
0x125
\0
0x126


0x123
s
0x123
t
h
0x123
i
0x124
!
0x125
\0
0x126

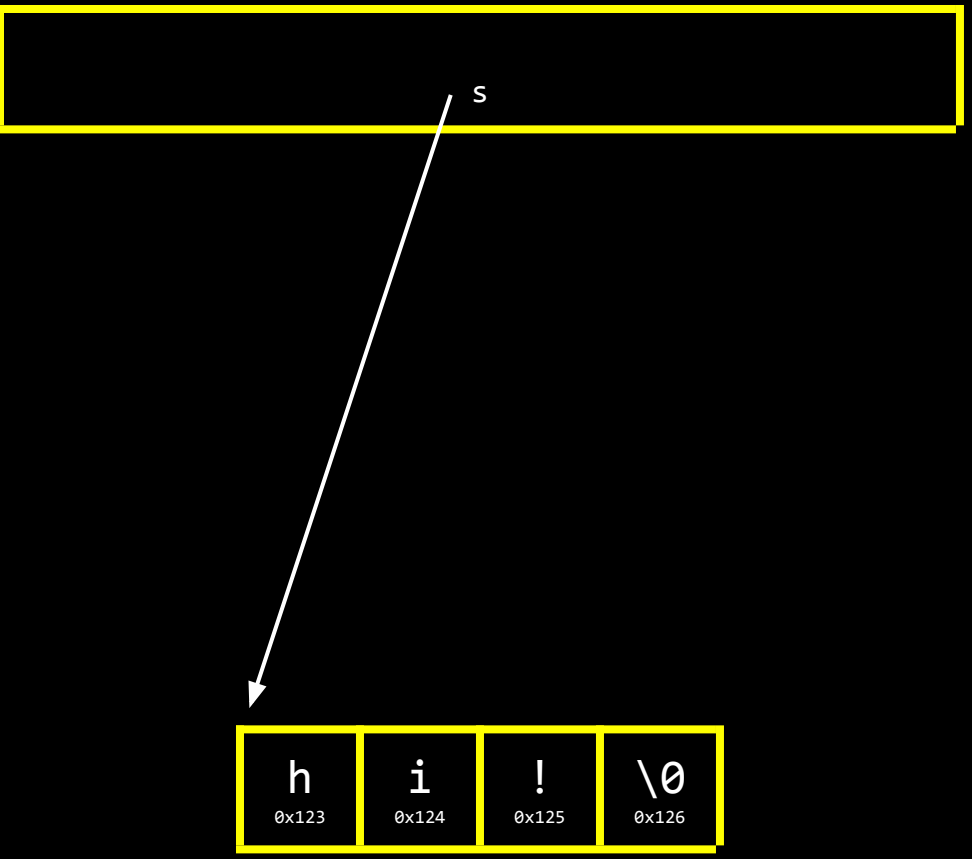

s
t
h
0x123
i
0x124
!
0x125
\0
0x126

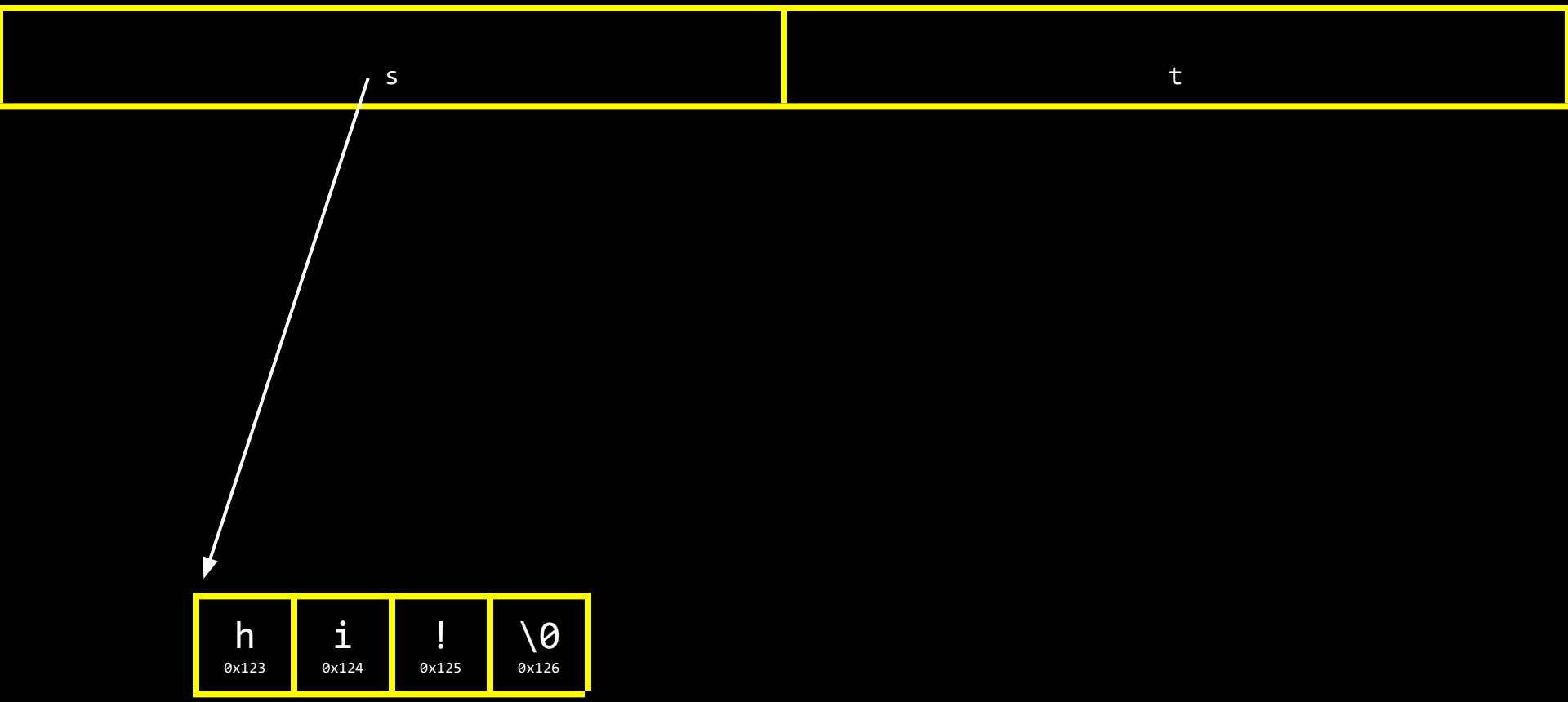

s
t
H
0x123
i
0x124
!
0x125
\0
0x126

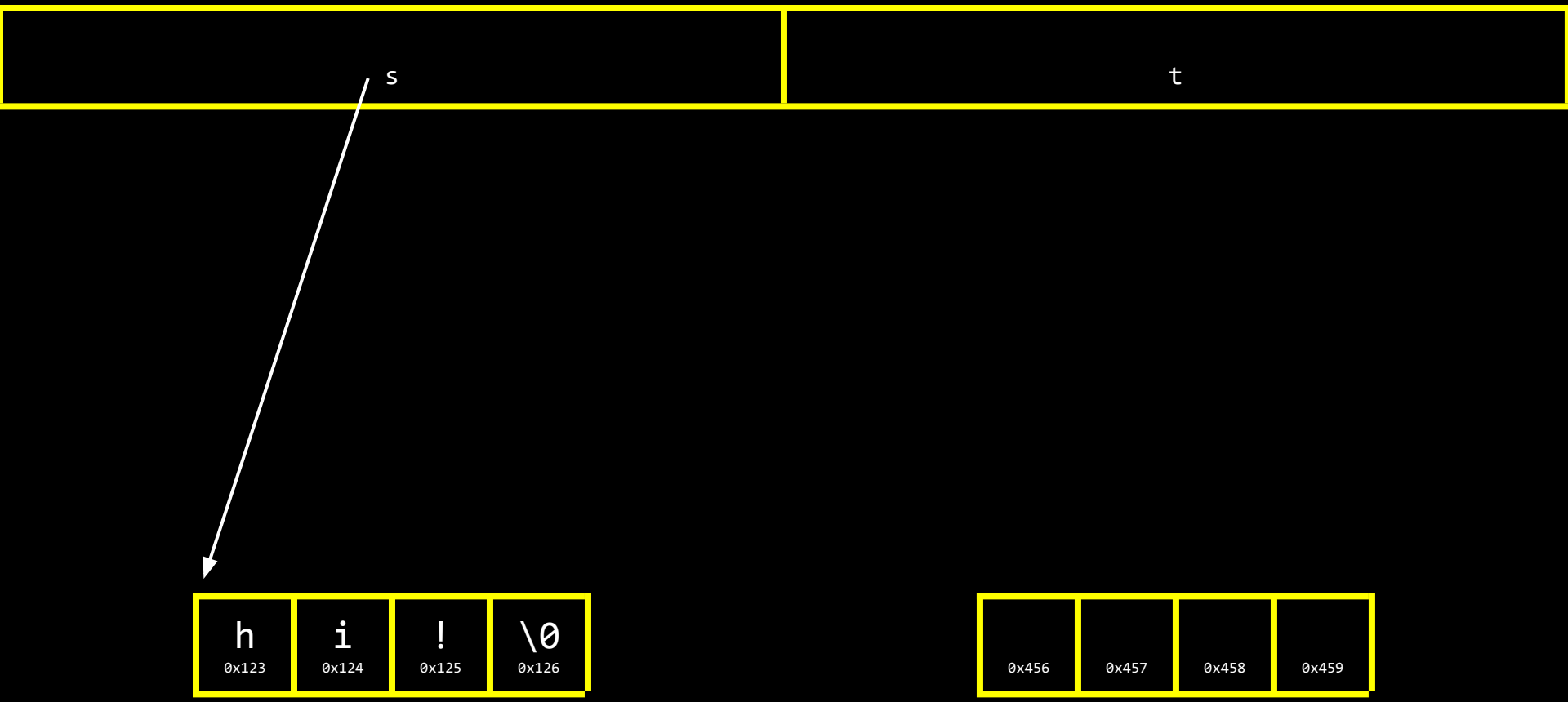
malloc

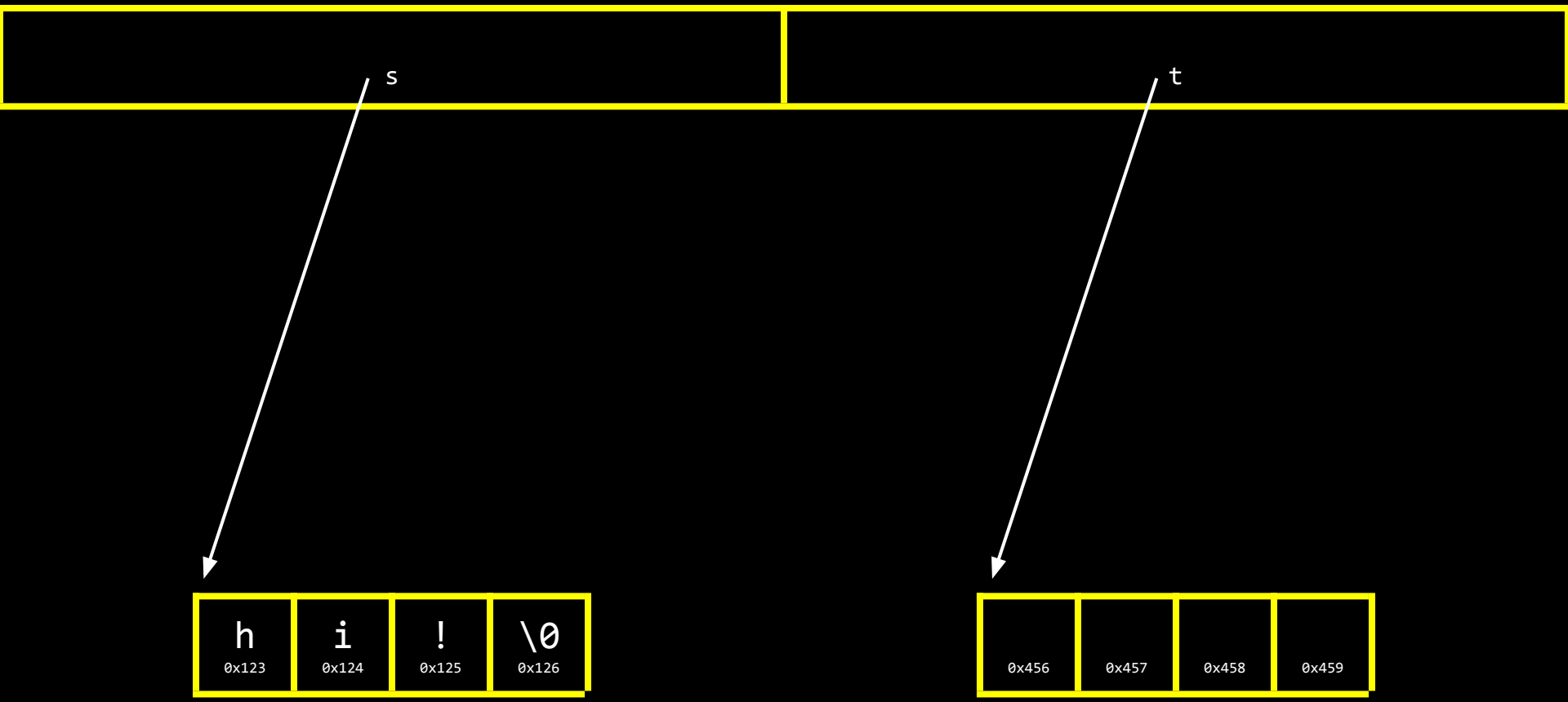
free

…


s
h
0x123
i
0x124
!
0x125
\0
0x126


s
t
h
0x123
i
0x124
!
0x125
\0
0x126


s
t
h
0x123
i
0x124
!
0x125
\0
0x126
0x456
0x457
0x458
0x459


s
t
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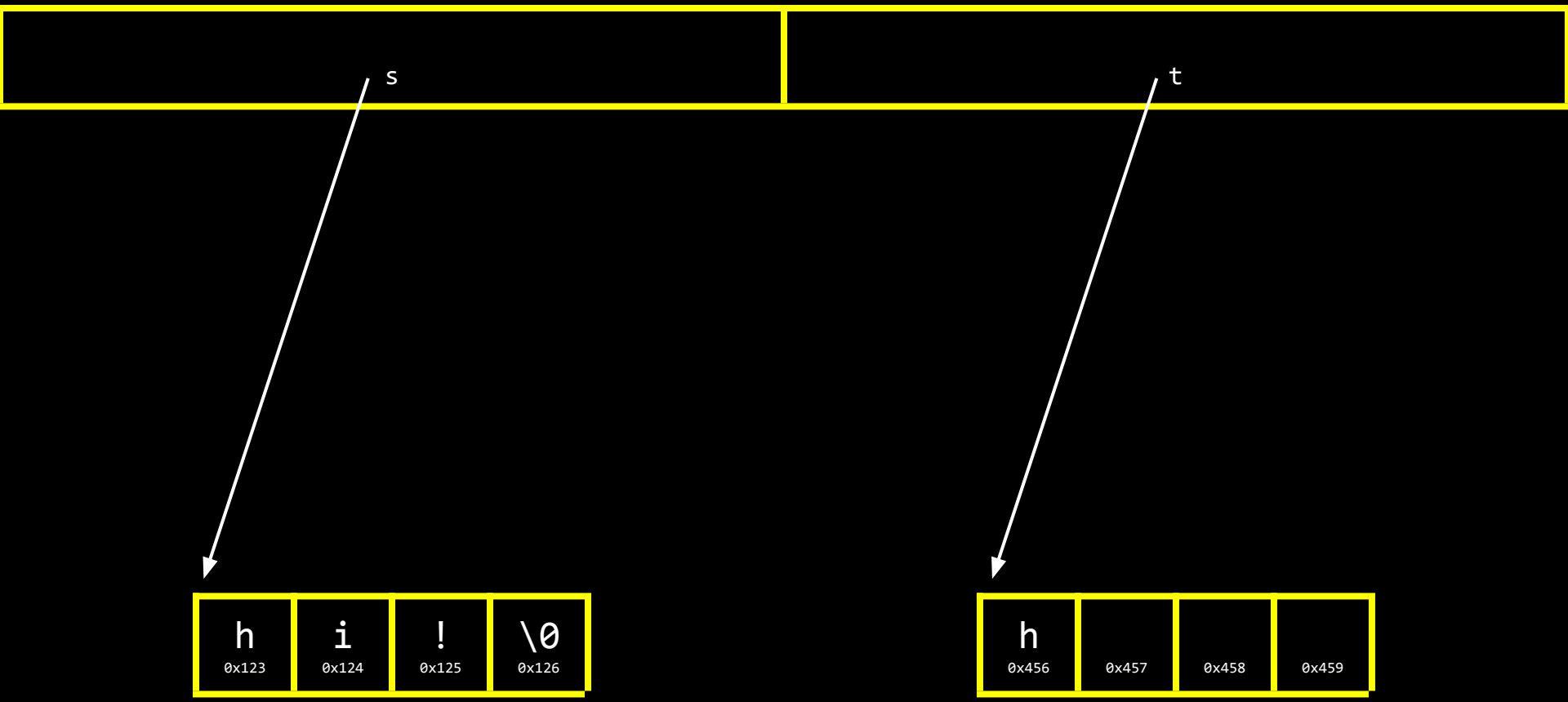
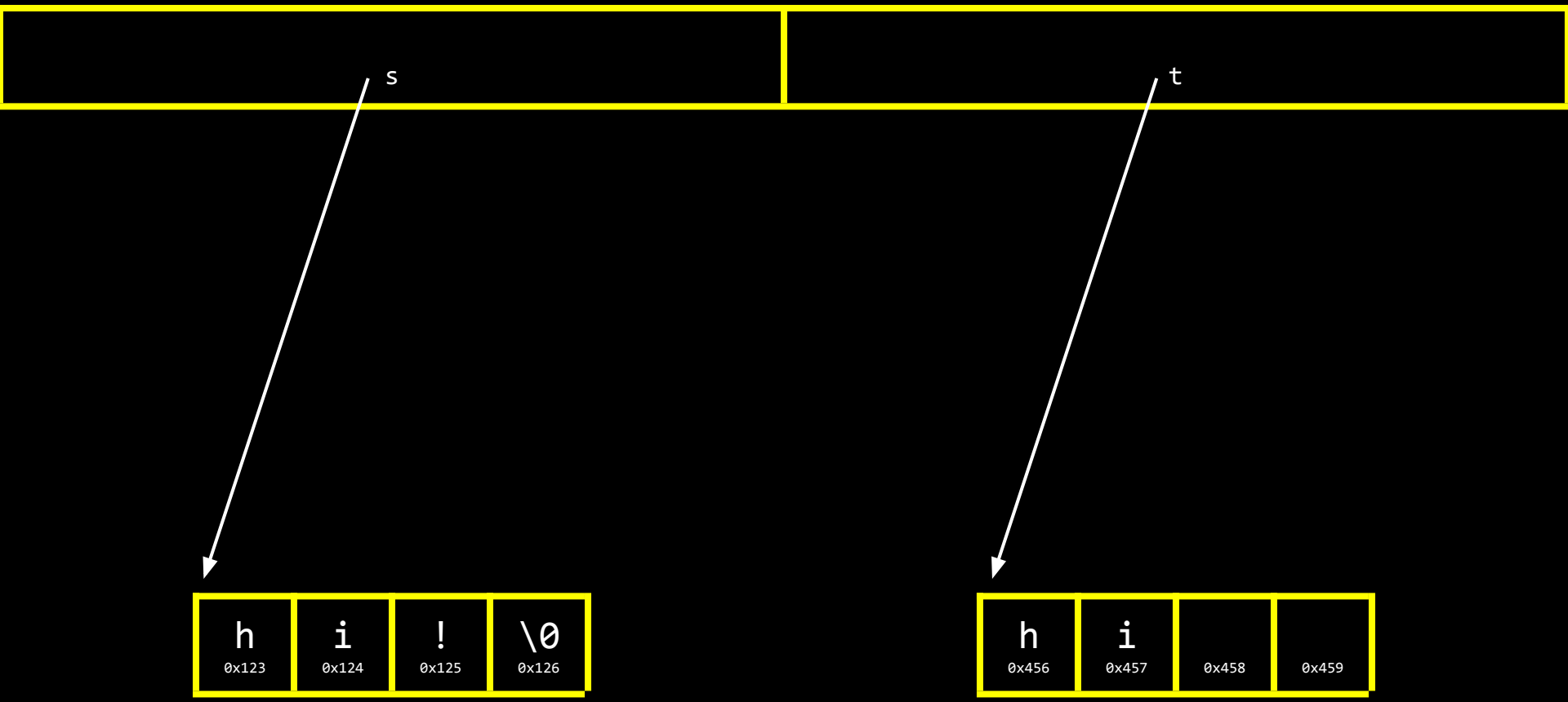
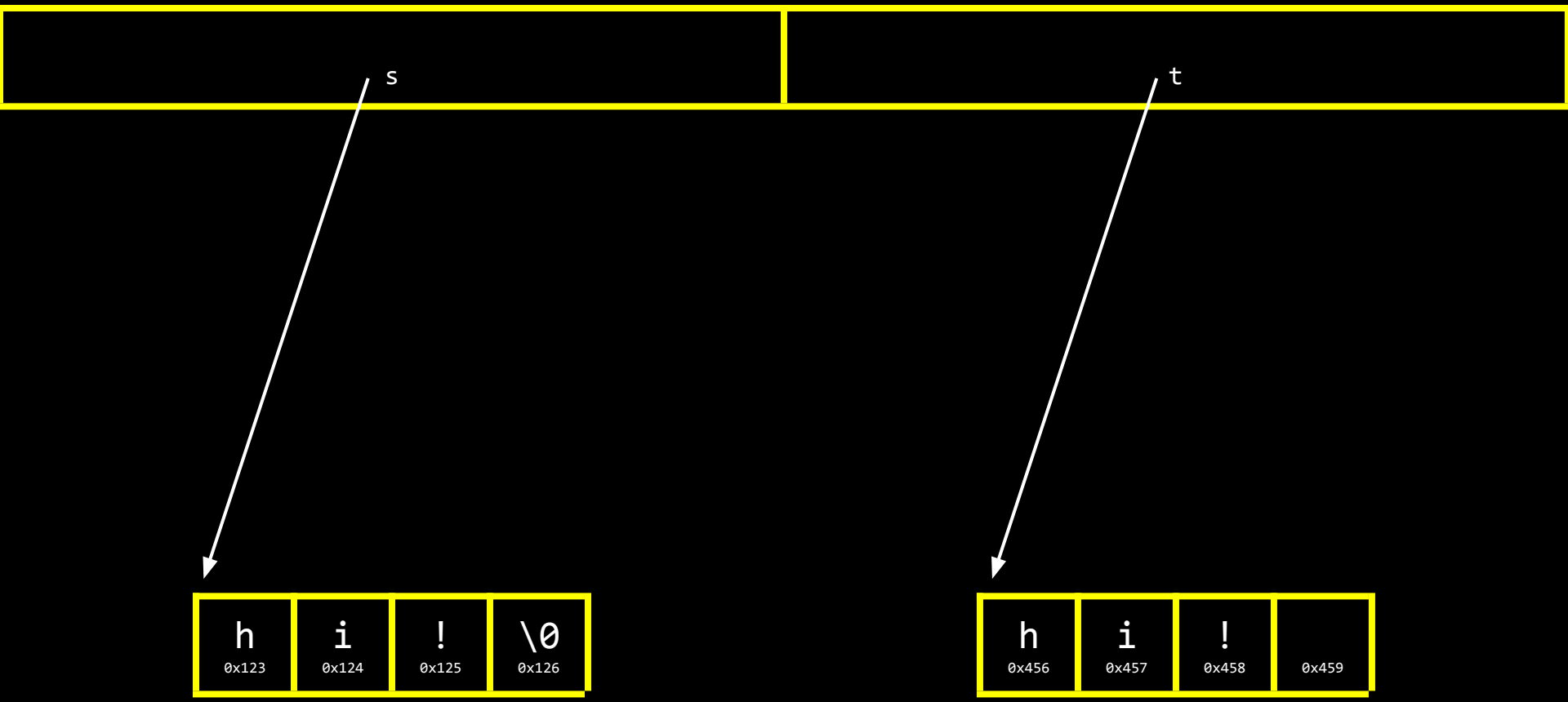
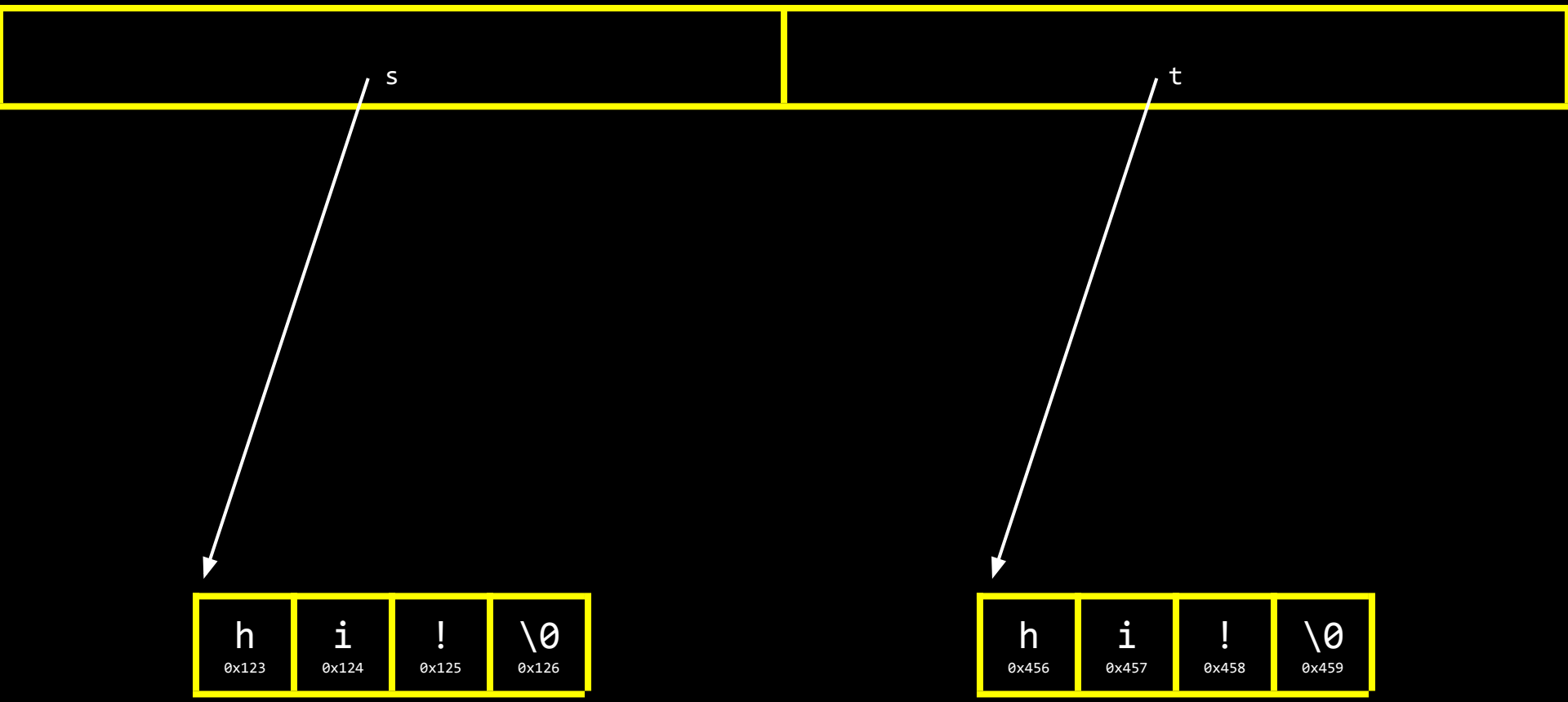
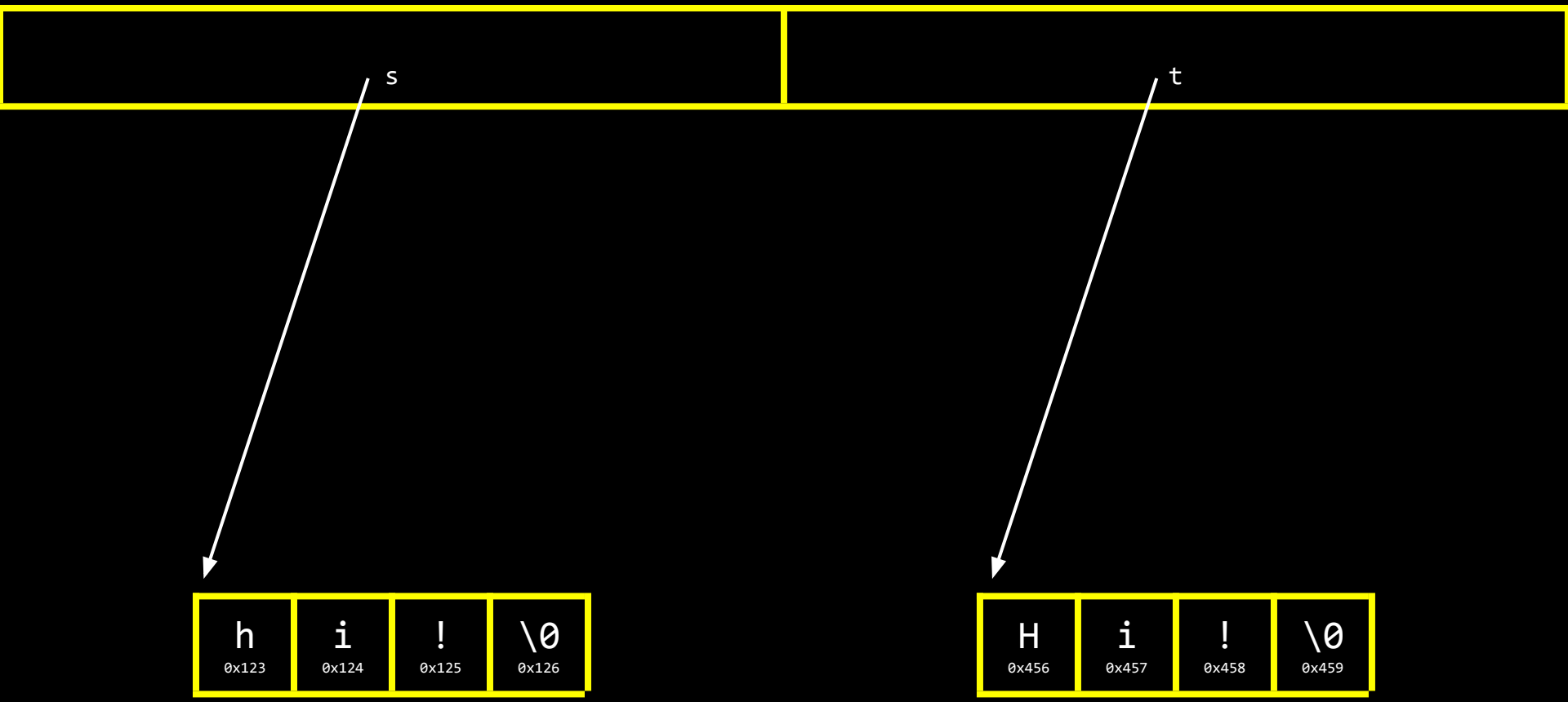
0x123
i
0x124
!
0x125
\0
0x126
0x456
0x457
0x458
0x459


s
t
h
0x123
i
0x124
!
0x125
\0
0x126
h
0x456
0x457
0x458
0x459


s
t
h
0x123
i
0x124
!
0x125
\0
0x126
h
0x456
i
0x457
0x458
0x459


s
t
h
0x123
i
0x124
!
0x125
\0
0x126
h
0x456
i
0x457
!
0x458
0x459


s
t
h
0x123
i
0x124
!
0x125
\0
0x126
h
0x456
i
0x457
!
0x458
\0
0x459


s
t
h
0x123
i
0x124
!
0x125
\0
0x126
H
0x456
i
0x457
!
0x458
\0
0x459

NULL

```
*y = 13;
```

valgrind

garbage values

```
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;
    *y = 13;

    y = x;

    *y = 13;
}
```

```
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;
    *y = 13;

    y = x;

    *y = 13;
}
```

```c
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;
    *y = 13;

    y = x;

    *y = 13;
}
```

```c
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;
    *y = 13;

    y = x;

    *y = 13;
}
```

```c
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;
    *y = 13;

    y = x;

    *y = 13;
}
```

```
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;


    y = x;

    *y = 13;
}
```

```
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;


    y = x;

    *y = 13;
}
```

```
int main(void)
{
    int *x;
    int *y;

    x = malloc(sizeof(int));

    *x = 42;


    y = x;

    *y = 13;
}
```

```
*y = 13;
```

```
void swap(int a, int b)
{



}
```

```
void swap(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

```
void swap(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

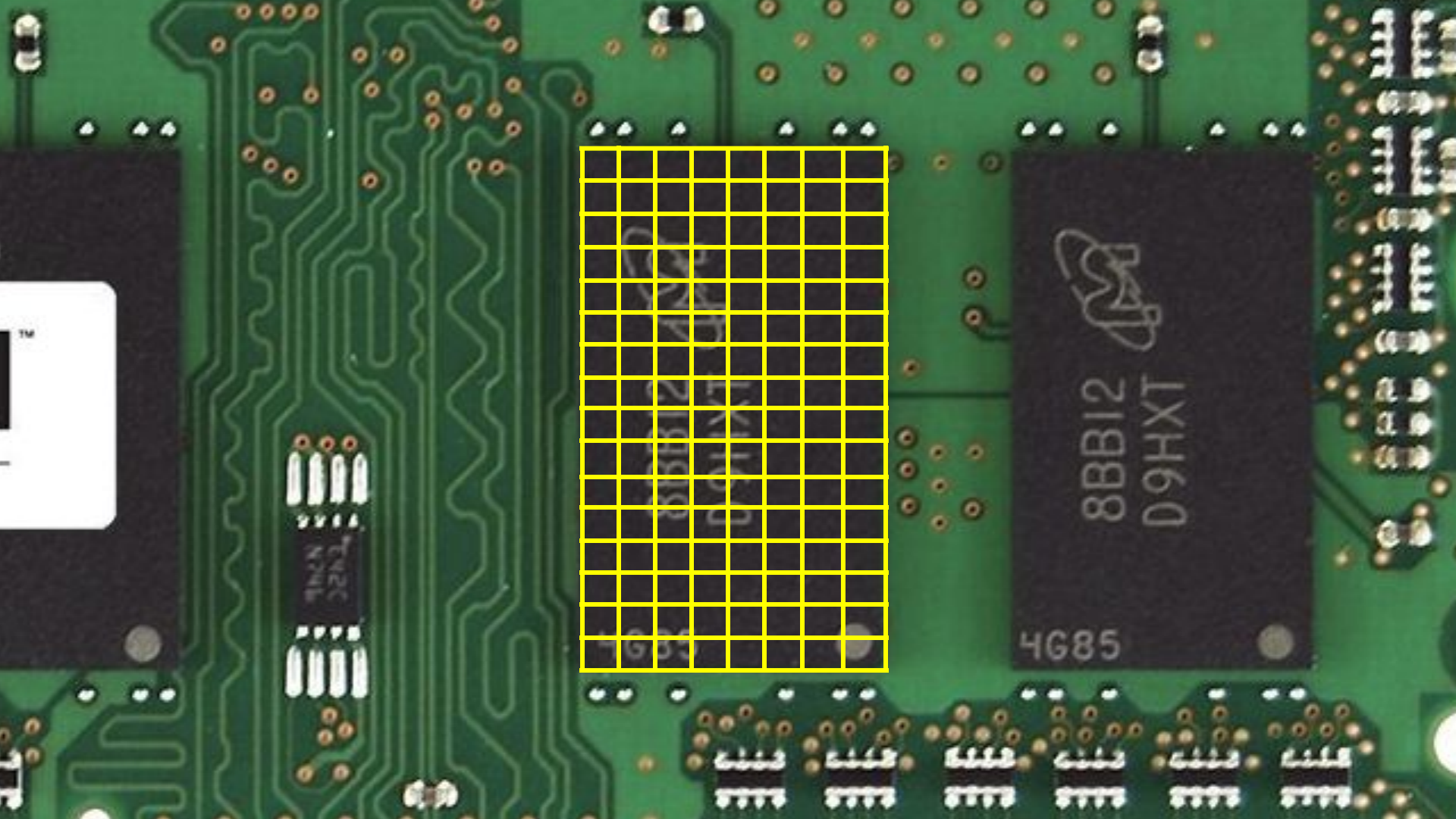
scope

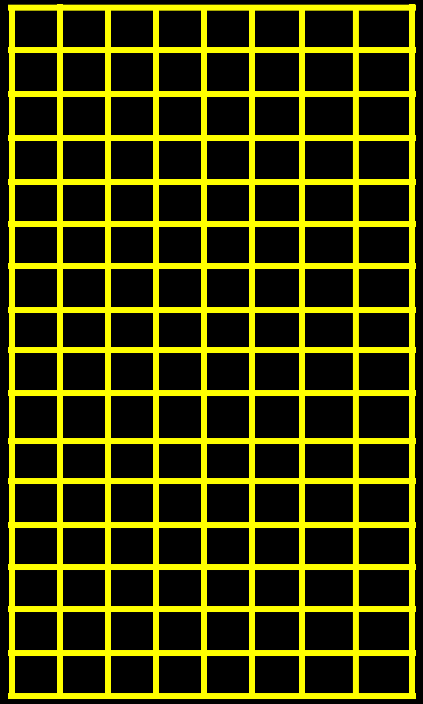
passing by value

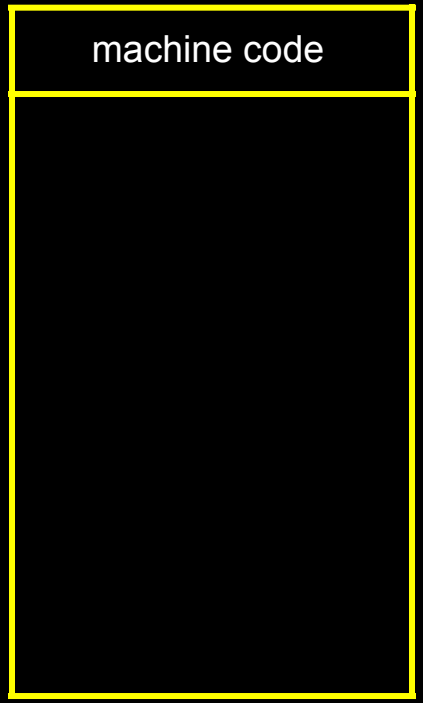
crucial
by Micron
47J6
4G85
8BB12
D9HXT
4G85
8BB12
D9HXT
4G85

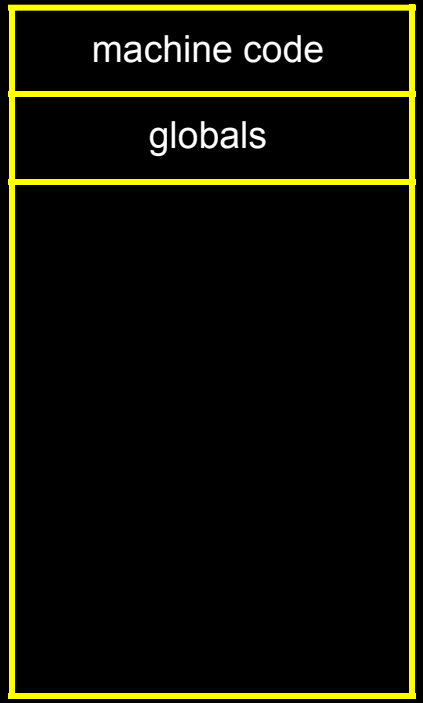
amazon.com

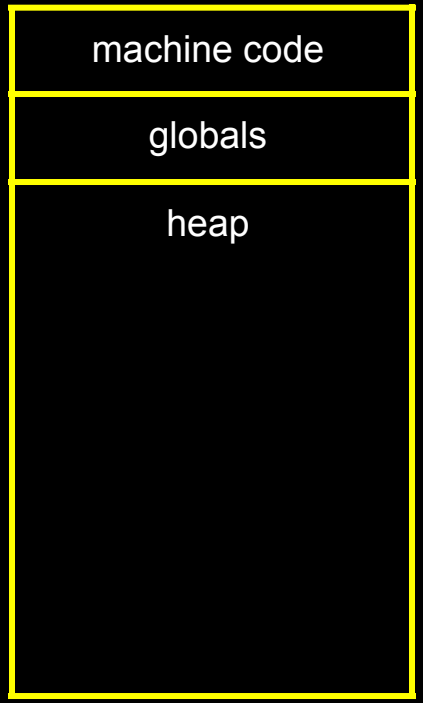
8BBI2
D9HXT
4G85
8BBI2
D9HXT
4G85

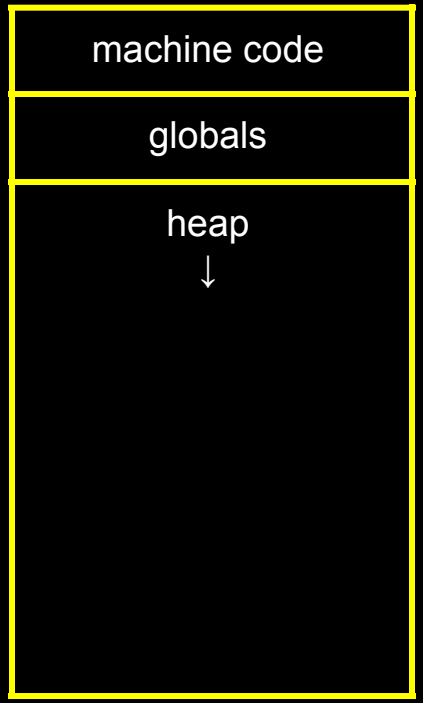

8BB12
D9HXT
4G85

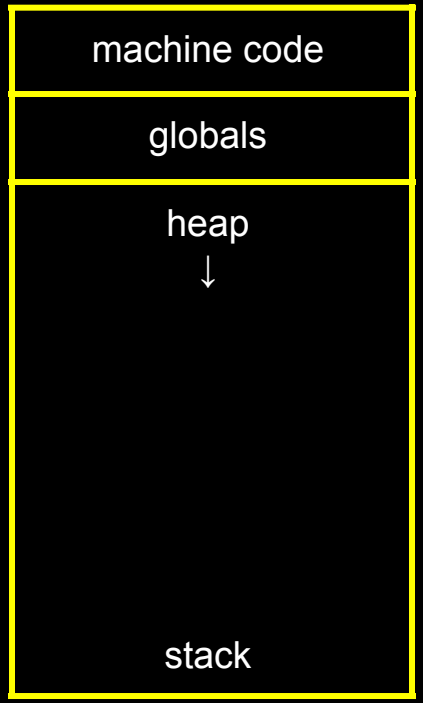


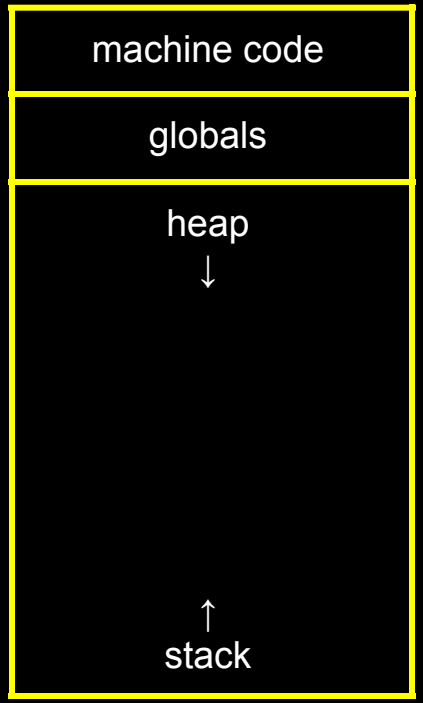

machine code

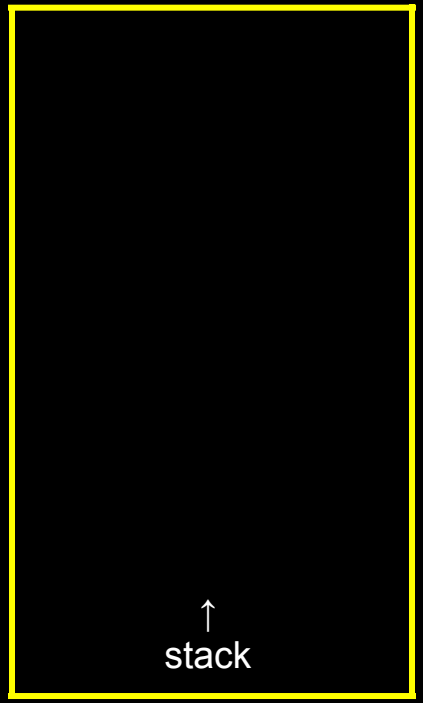

machine code
globals


machine code
globals
heap

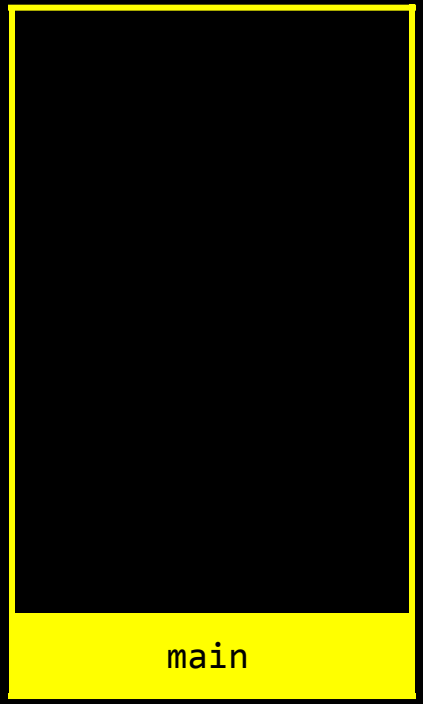

machine code
globals
heap
↓

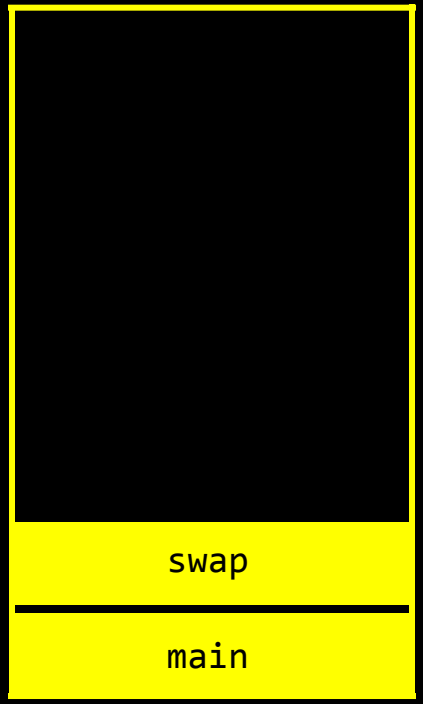

machine code
globals
heap
↓
stack

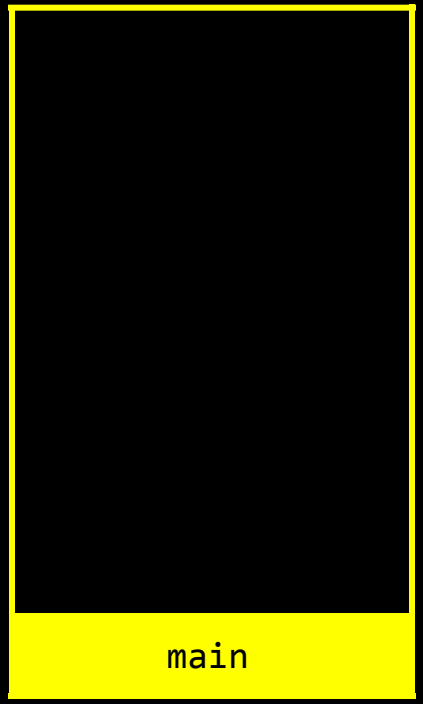

machine code
globals
heap
↓
↑
stack

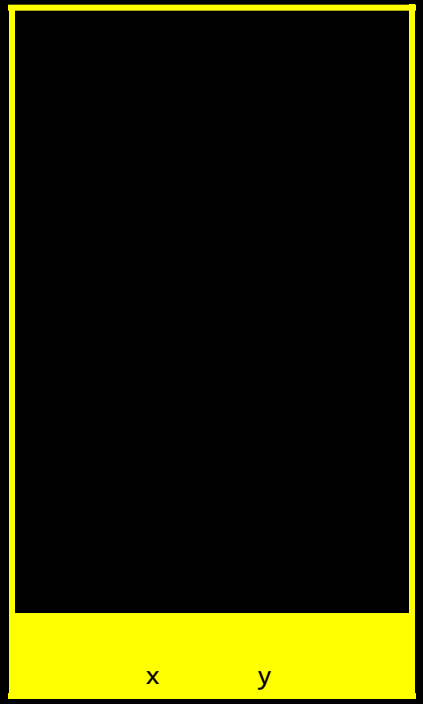

↑
stack

```
void swap(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

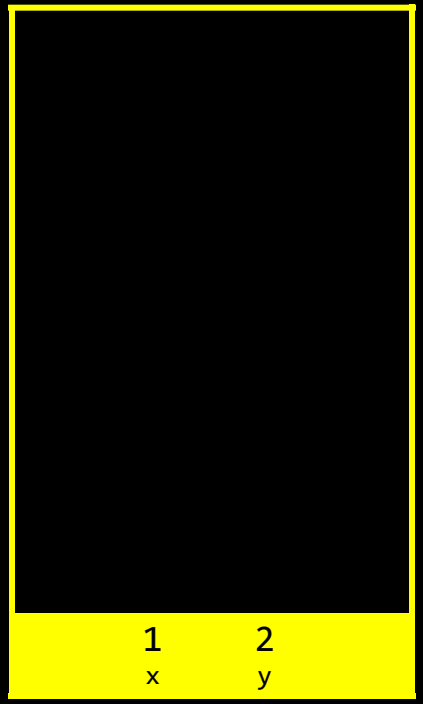

main


swap
main

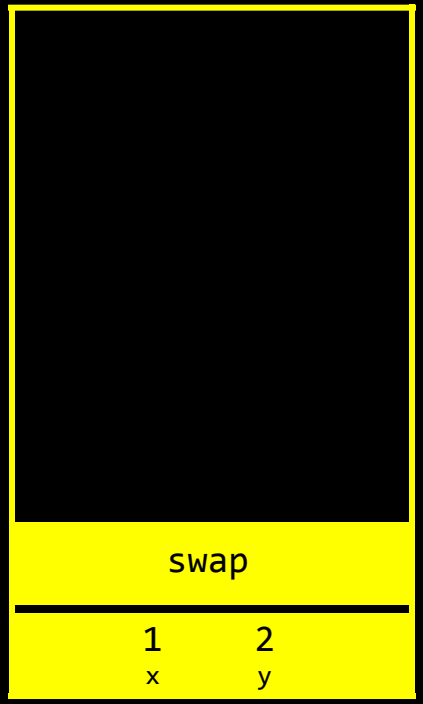

main


x
y

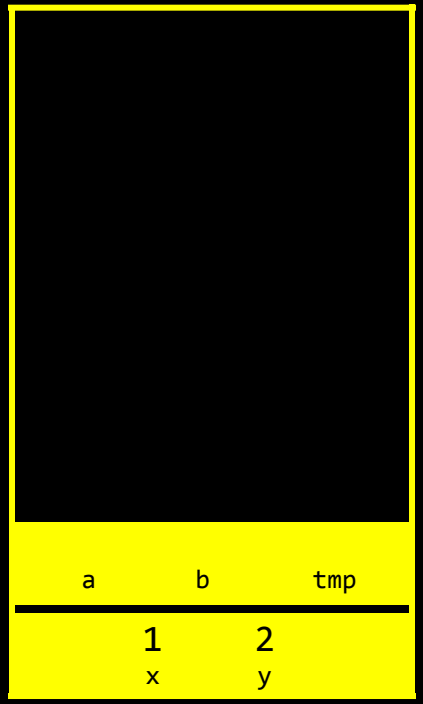
```
int x = 1;
int y = 2;
```



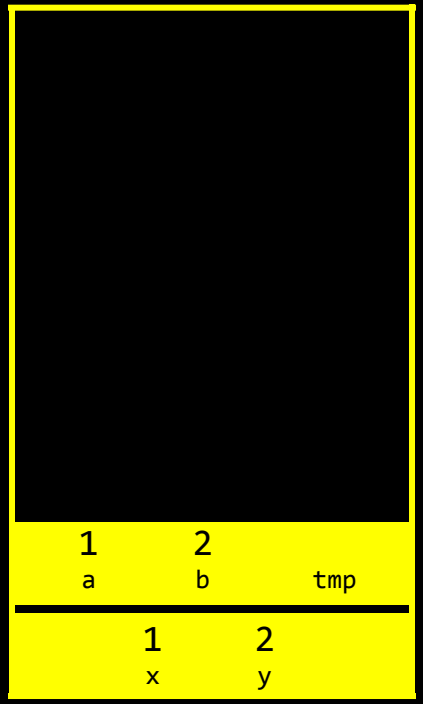
```
swap(x, y);
```



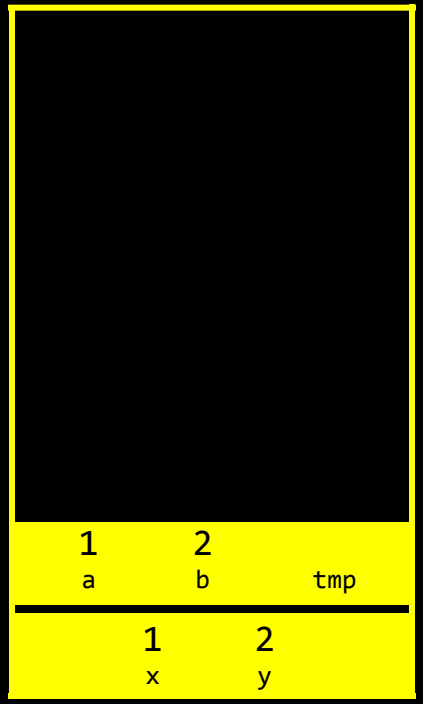
```
swap(x, y);
```



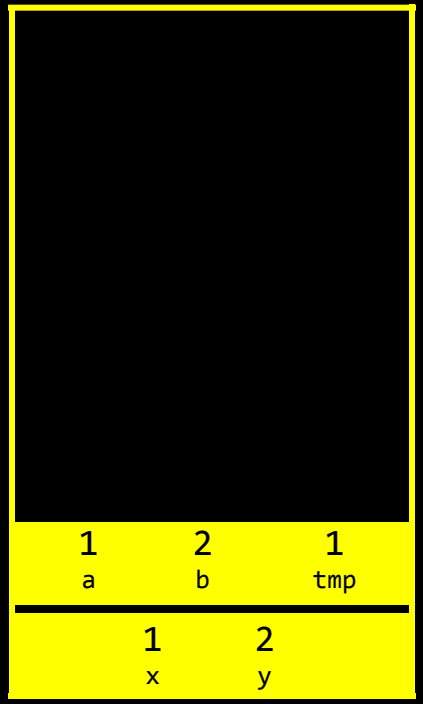
```
void swap(int a, int b);
```



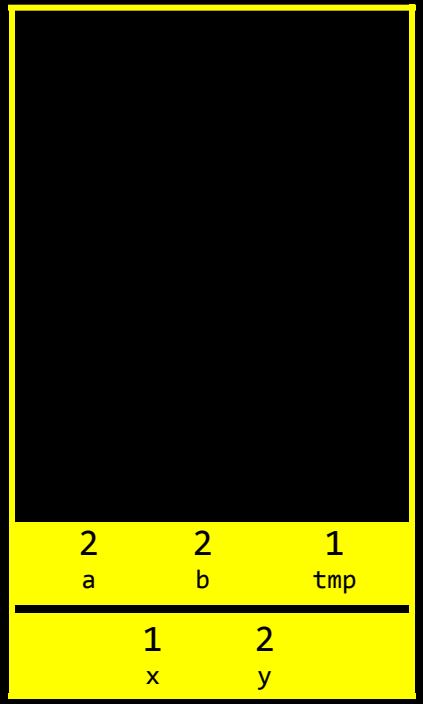
```
int tmp = a;
a = b;
b = tmp;
```



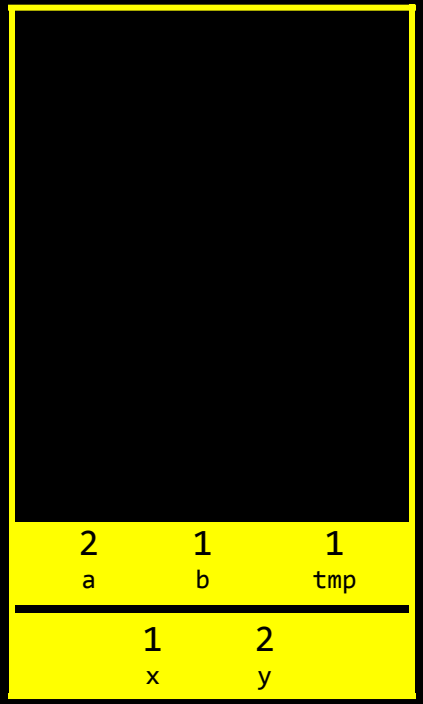
```
int tmp = a;
a = b;
b = tmp;
```

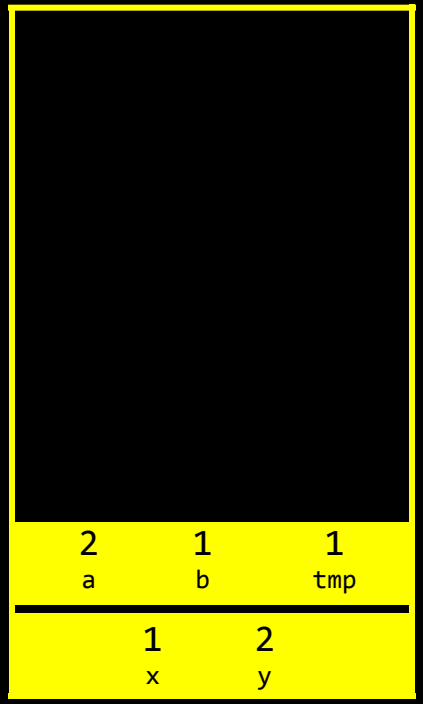


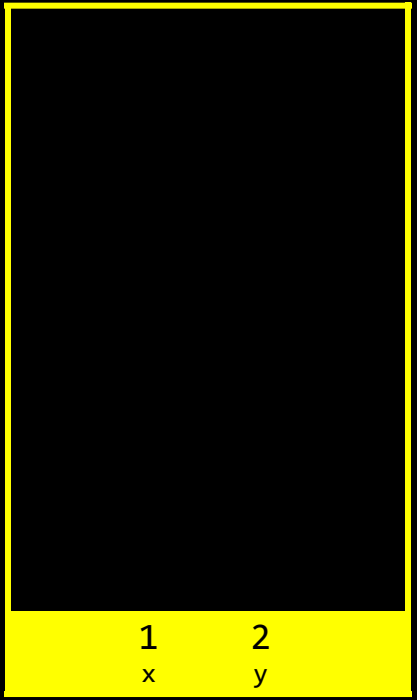
```
int tmp = a;
a = b;
b = tmp;
```



```
int tmp = a;
a = b;
b = tmp;
```




2
a
1
b
1
tmp
1
x
2
y

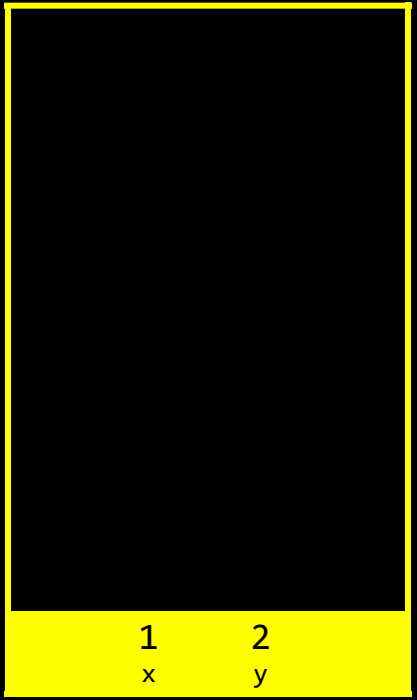

1
x
2
y

passing by reference

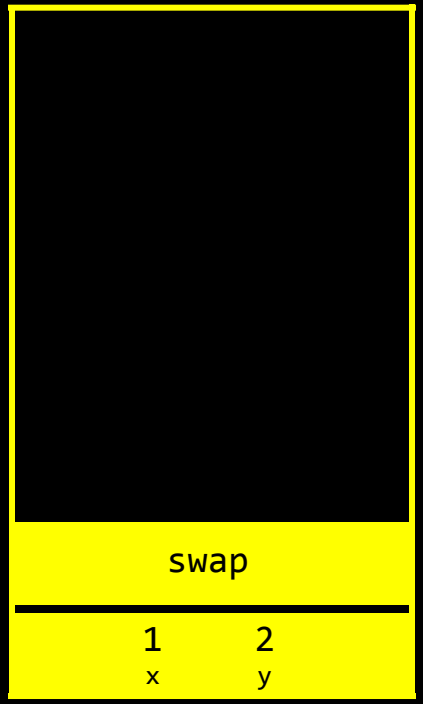
```
void swap(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

```
void swap(int *a, int *b)
{
    int tmp = *a;
    *a = *b;
    *b = tmp;
}
```

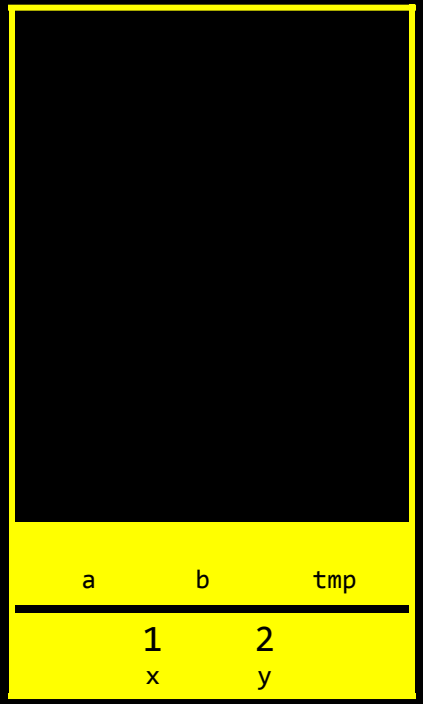
```
void swap(int *a, int *b)
{
    int tmp = *a;
    *a = *b;
    *b = tmp;
}
```


1
x
2
y

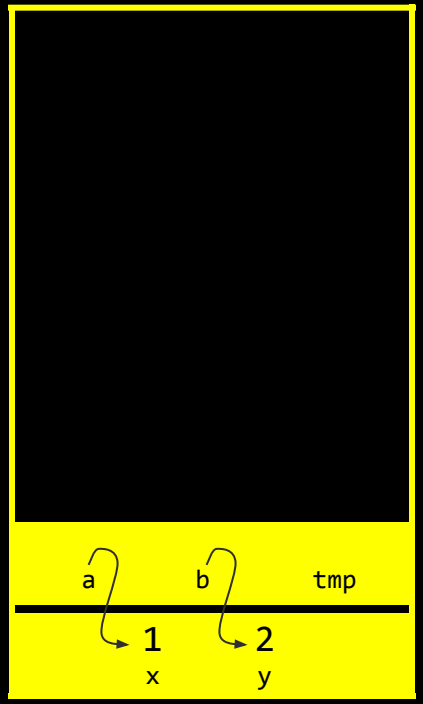
```
swap(x, y);
```



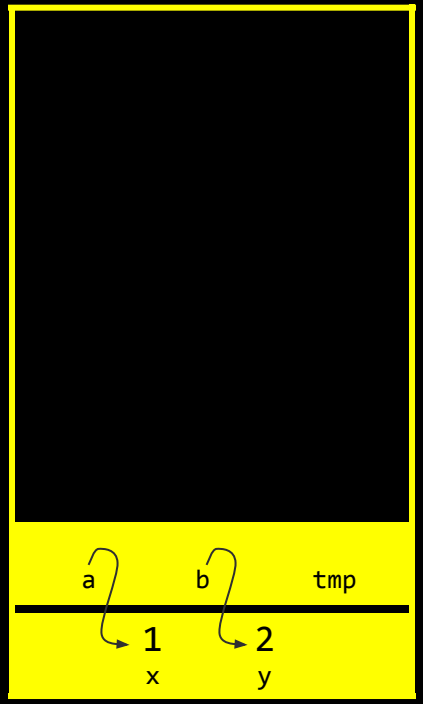
```
swap(x, y);
```



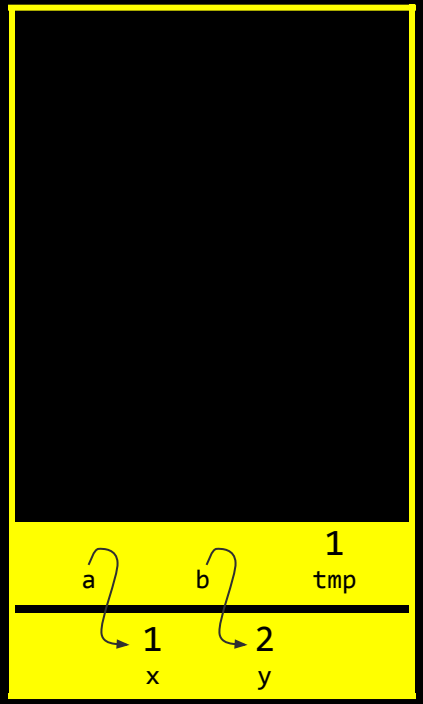
```
void swap(int *a, int *b);
```



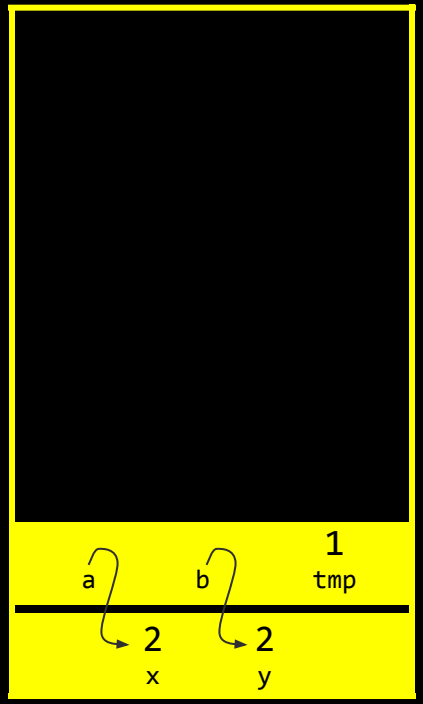
```
int tmp = *a;
*a = *b;
*b = tmp;
```



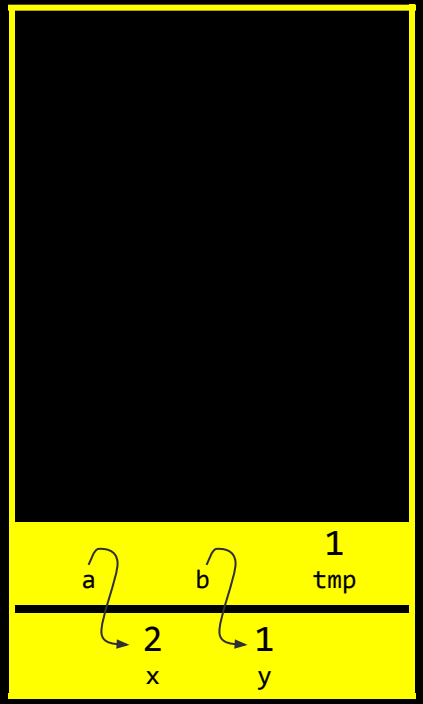
```
int tmp = *a;
*a = *b;
*b = tmp;
```

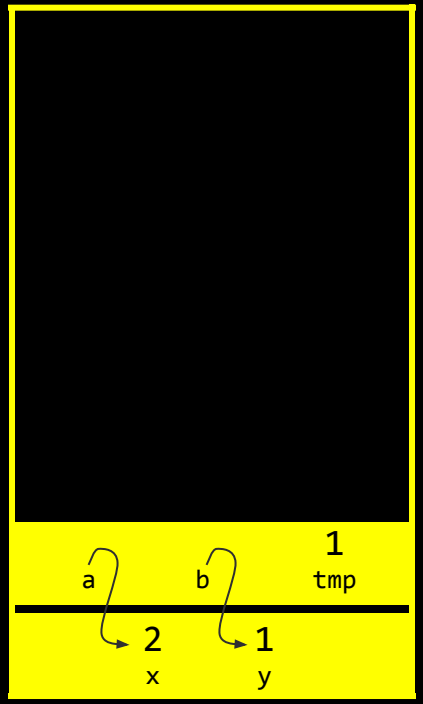


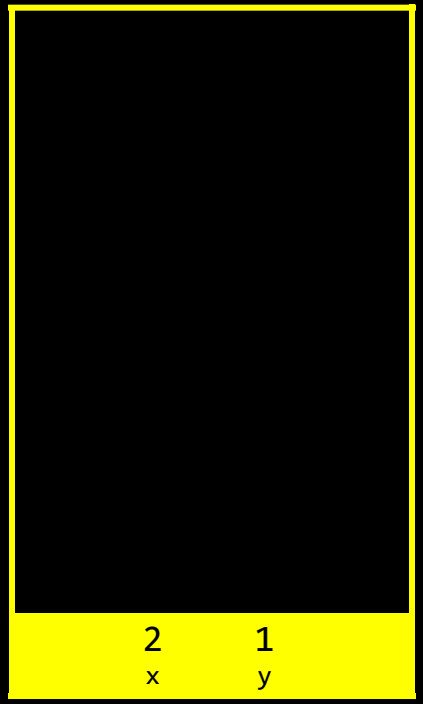
```
int tmp = *a;
*a = *b;
*b = tmp;
```



```
int tmp = *a;
*a = *b;
*b = tmp;
```

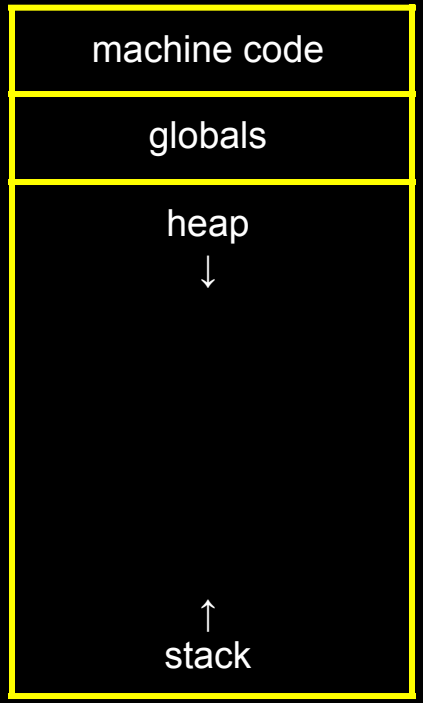


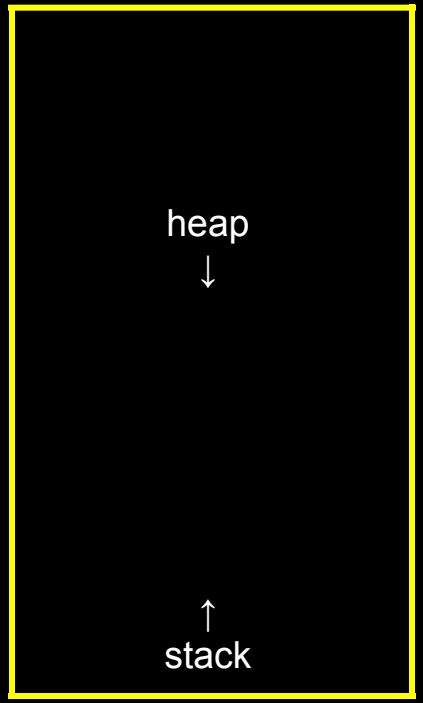

1
tmp
a
b
2
x
1
y


2
x
1
y

```
void swap(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

```
void swap(int *a, int *b)
{
    int tmp = *a;
    *a = *b;
    *b = tmp;
}
```


machine code
globals
heap
↓
↑
stack


heap
↓
↑
stack

heap overflow

stack overflow

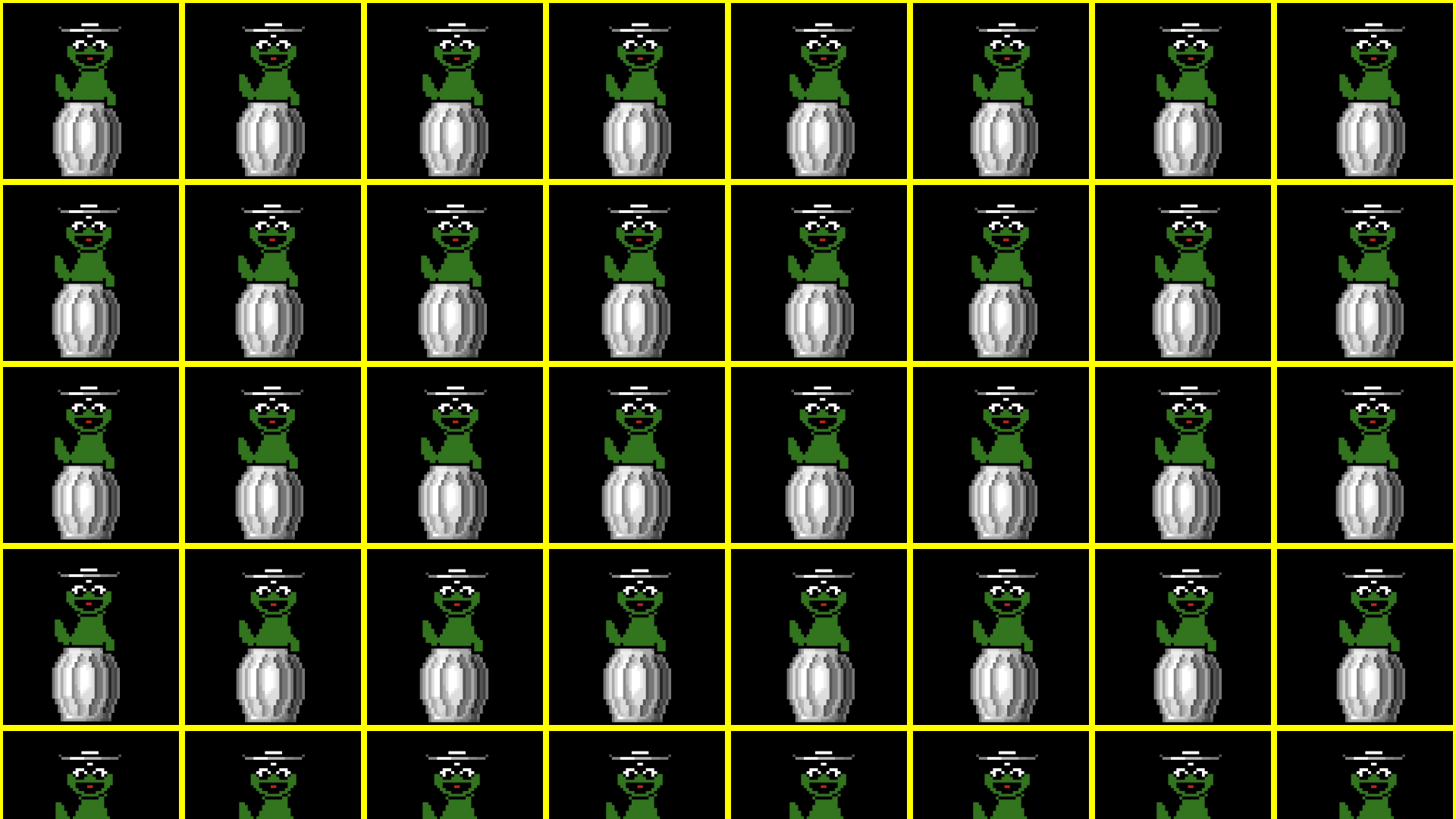
buffer overflow

```
get_char

get_double

get_float

get_int

get_long

get_string

...
```

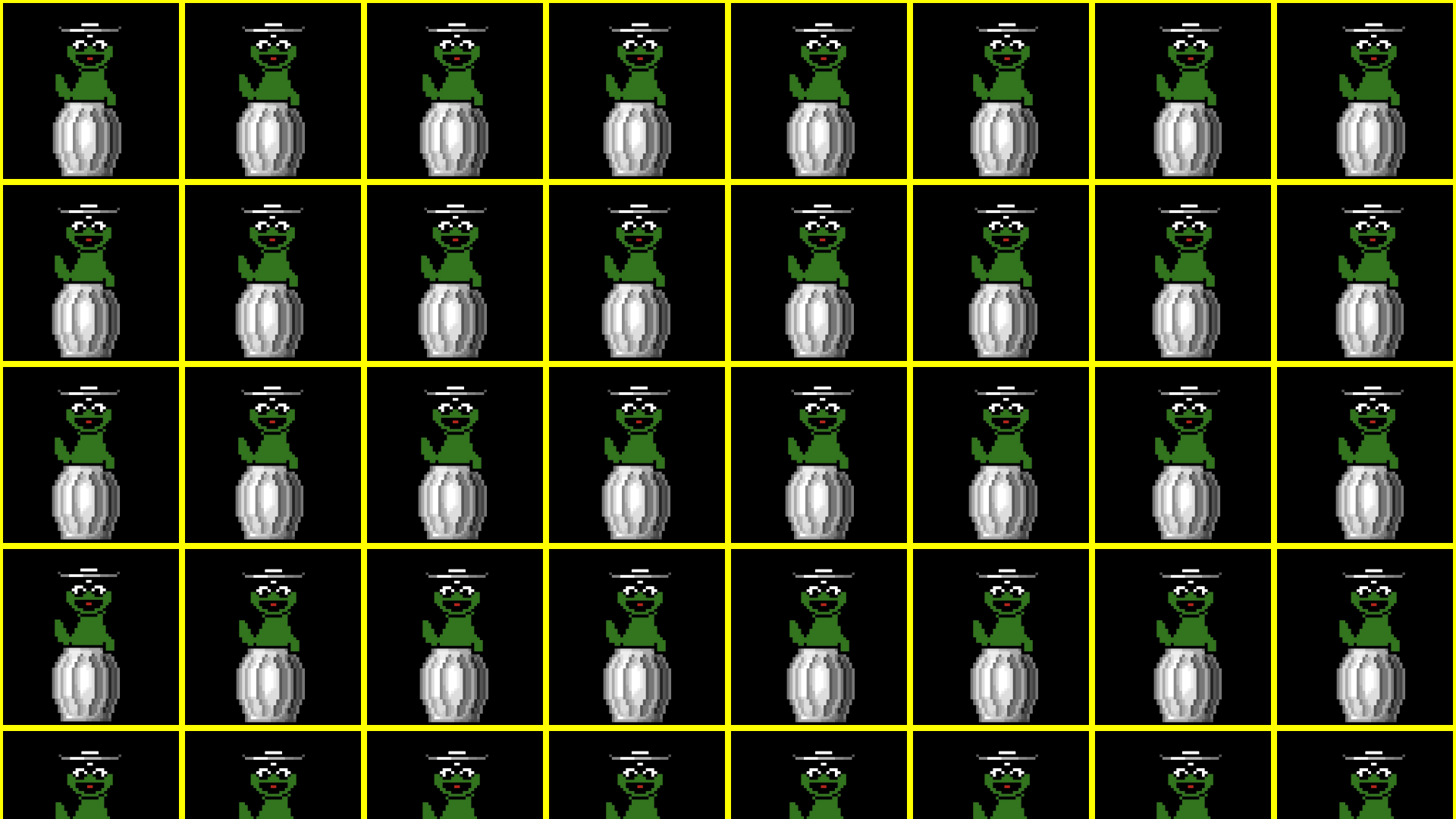
get_int

scanf

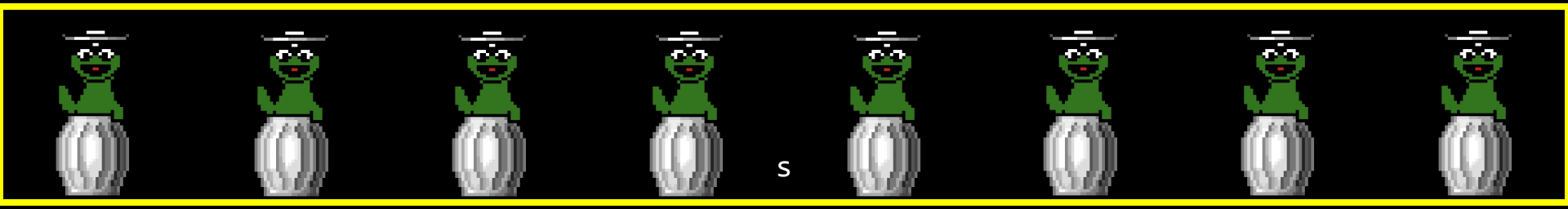
...



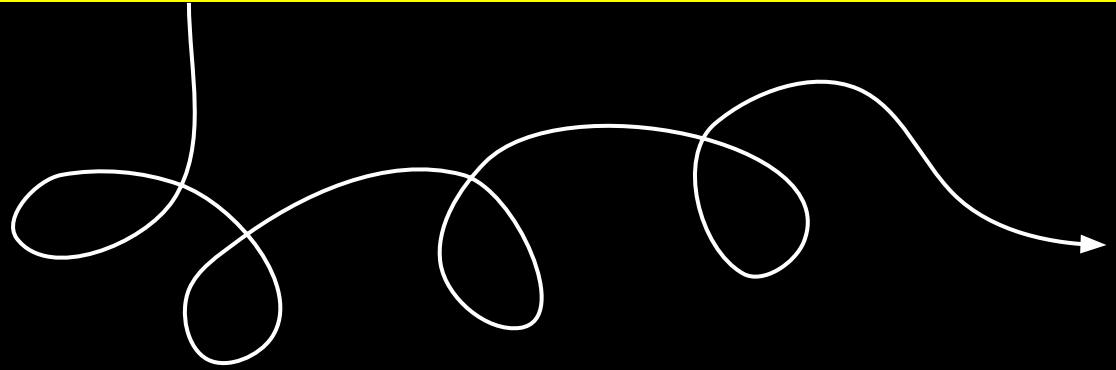
50
n

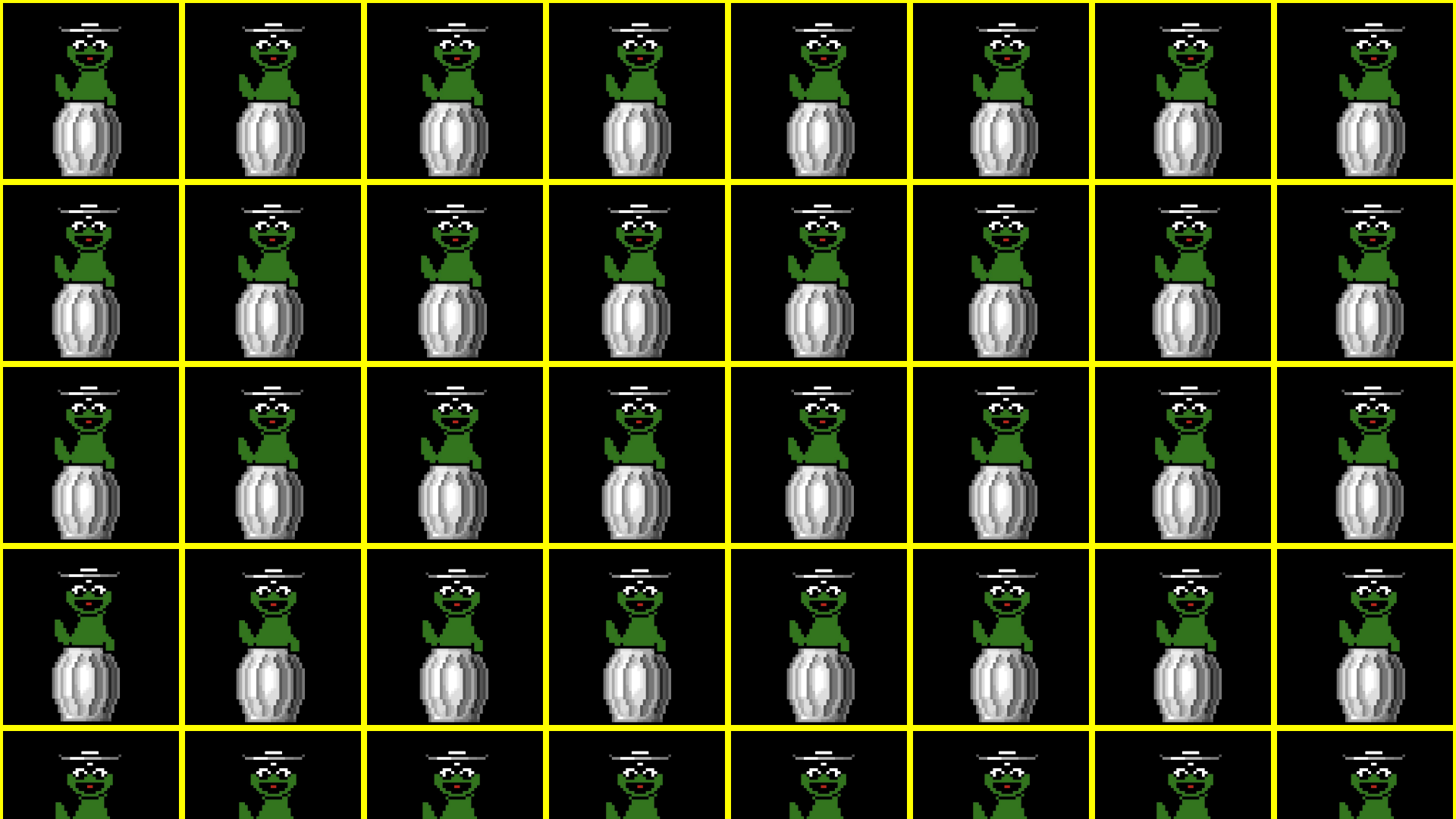


S


S

s





S

file I/O

fopen

fclose

fprintf

fscanf

fread

fwrite

fseek

…

BMP

MAN, I SUCK AT THIS GAME. CAN YOU GIVE ME A FEW POINTERS?
0x3A28213A 0x6339392C, 0x7363682E.
I HATE YOU.

This is CS50